

**Inventory Management and Asset Tracking System using barcode for
Global Reciprocal Colleges General Services Department**

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In Partial Fulfillment of the requirement
For the Subject CAPS1: Capstone Study

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APPROVAL SHEET

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ABSTRACT

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CHAPTER I

THE PROBLEM AND ITS BACKGROUND

This chapter presents the context of our project, the objective of this study, the scope and limitation of the study, significances of the study and definition of key terms.

1.1 PROJECT CONTEXT

In today's world, technology plays a critical role. It has become an integral part of our daily lives, and it has made our lives easier, more efficient, and more convenient as a result. Technology has also allowed us to have more access to information than ever before with less effort and in less time. Because of these advancements, technology has enabled us to achieve things that were once thought impossible and has created new opportunities for growth and development. Nonetheless, the impact of technology in our world today can never be overstated. Because of this, automated systems have replaced paper-based systems in every company and corporate organization. It now allows the companies to streamline their processes and keep track of their data effectively and accurately.

Inventory management has been transformed by technology, making it into a automated one from previously manual operations. We all know that manual inventory can result in difficulties in keeping track of the items, as well as the inventory levels, locations, and movements, which can lead to errors such as miscounting, incorrect data entry, and misplacement of inventory items. These errors can also lead to inaccurate inventory levels, which can result in stockouts, overstocking, and the loss of items. Also, in the event of a fire or other disaster, manual inventory management can also lead to significant losses if inventory records are destroyed or lost. Without a proper backup system, businesses and organizations can lose critical information regarding their data, making it difficult to resume operations quickly.

The General Services Department of Global Reciprocal Colleges is in charge of managing facilities, supplies, maintenance, and security. It is also accountable for the quotation and procurement of supplies and materials needed to maintain the school's facilities.

The department is still using a manual inventory for their assets, keeping their records in a pile of papers and an Excel spreadsheet. Which is inefficient for tracking and maintaining records of assets, especially when it comes to tracking the location and movement of their assets. The problems experienced by the department are basically data redundancy, unorganized records, or missing inventory records. They also stated that there are situations where part of the reports is inaccurate, and there are instances where the items in their inventory keep disappearing for unknown reasons. They are also experiencing problems tracking their equipment, which sometimes is not returned on time.

So, by automating their inventory management, this would benefit them in a variety of ways. Furthermore, automated inventory management systems are often more accurate than manual methods and can help the department avoid costly mistakes and losses due to overstocking or understocking. Therefore, creating a system that will automate the processes of inventory management in the General Services Department can help them track their inventory, monitor stock levels, track their assets, and generate reports on the inventory's transactions. Not only that, the system will also have a barcode scanner to lessen their work on manually checking the items that are checking in and checking out. Also, there will be a report module that will be stored on the system database for their data on paper, Excel reports, and quotations that they are keeping in folders around their department room.

1.2 OBJECTIVES OF THE STUDY

GENERAL OBJECTIVE

The general objective of the study is to design and develop an Inventory Management and Equipment Tracking System using barcode scanner for Global Reciprocal Colleges General Services Department and automate their current process. The system features different modules for the staff to use. This includes inventory, equipment monitoring, & procurement process.

SPECIFIC OBJECTIVE

- To keep track all assets of the General Services Department and improve their inventory management.
- To lessen the existing problems of the General Services Department inventory management.
- To keep track of the status of the procurement whether they are approved, pending, and cancelled.
- A system that provides accurate reports and information.
- Increases the department's overall performance and improves efficiency.

1.3 SCOPE AND LIMITATIONS OF THE STUDY

SCOPE

This study focuses on the development of Inventory Management and Asset Tracking System using barcode for Global Reciprocal Colleges General Services Department to improve the current process mostly storing and filing important records.

- a. The system allows multiple users to access the system mainly the department head, the building admin, and the inventory admin.

- The department head is the only authorized person who can create, edit, delete record, update a user account, and set a specific permission or access rights. The department head can monitor all the modules of the system, additionally he can modify or update and inspect all modules. Other modules that is exclusive only for the department head's account are maintenance, user logs and audit trail.
- The maintenance module is where he can add, update, and remove information in all other modules. This is also where he can alter privileges of a user, create another user, or disable some of them.
- The user logs are where he can monitor log history of users.
- And the audit trail module is where he can track major events made by a user like adding, updating, or removing information from the system and to monitor the information entered in the system too.
- b. The system can generate summary reports of all transactions happening in the system.
 - This includes the average number of items and equipment every department must get.
 - And quotations at purchase order to know the step – by – step status of items that are to be deliver and delivered.
- c. The system also uses a barcode scanner for tracking of borrowed equipment and consumable item stocks out.
 - The generated barcodes are then printed onto labels or tags, which are attached to the respective items. These labels contain the encoded information per items in the inventory in the form of a series of black, red, blue, white bars etc. When an item needs to be borrowed or its stock is checked out, the barcode is scanned using a barcode scanner.
 - The barcode scanner captures the information encoded in the barcode, such as the item's unique identifier or stock details. Then the scanned data is sent to a central database or inventory management system.

- This system maintains a record of borrowed equipment, consumable item stock levels, and associated information. After the items are out the system updates the status of the borrowed equipment or adjusted stock levels based on the scanned data. It can provide real-time information on item availability, due dates, and generate alerts for overdue returns or low stock levels.
- d. The system is also LAN-based because the department has three personnel that will use the system. Because the system is LAN-based this would provide local connectivity, allowing devices within the LAN to communicate with each other. This enables the sharing of resources, such as files.

LIMITATIONS

The study will not include the other operations that the General Services Department manages such as the process of canvassing, security process such as INGRESS, EGRESS and Gate Pass. And will only focus on automating their inventory management, tracking of items, and compiling of completed purchase order of the General Services Department.

The other limitations of the study are that the system is only accessible on a desktop, other devices such as phones and tablets does not have access to the system. Also, the process of borrowing of items and equipments are limited only to the current students, professors, and departments only of Global Reciprocal Colleges.

1.4 SIGNIFICANCE OF THE STUDY

This part of the study explained the importance of the development of the system to the following individuals:

GLOBAL RECIPROCAL COLLEGES

The primary benefactor of this study is the school of Global Reciprocal Colleges. The study on implementing a barcode scanning system for tracking borrowed equipment and consumable item stocks would primarily benefit the school's asset management processes.

GENERAL SERVICES DEPARTMENT

The next benefactor of this study is the General Services Department of Global Reciprocal Colleges. The system will automate their previous manual processes and will enable them to manage, organize, and store all purchase orders and reports with the system effeciently and accurately.

SCHOOL DEPARTMENTS

The other benefactor of this study are the departments of the Global Reciprocal Colleges. The system will easily determine if the department is already added on the system's school department module. And the system can provide them information before they borrow or request the items they needed.

PROPONENTS

This will serve them the first step towards to success in the field of expertise. This will also help them apply the knowledge they have gained throughout the project.

FUTURE PROPONENTS

The proposed study will serve as a reference for other or future proponents with the same study or system. The future proponents can base their systems from this study and may improve further the capability of the developed thesis project.

1.5 DEFINITION OF TERMS

Upon the implementation, the following terms that are not familiar to the users and future researchers will be the following, it has been defined clearly and conceptually for better understanding.

Asset – refers to a valuable resource or item owned by an individual or organization, typically with economic value. Assets can include physical items (e.g., equipment, vehicles, buildings) as well as intangible items (e.g., intellectual property, patents, trademarks).

Audit Trails – refers to a chronological record of activities, changes, or transactions within a system or process. Audit trails are used for tracking and documenting events, providing a history of actions, facilitating accountability, and supporting compliance or investigation purposes.

Building Admin – refers to individual or team responsible for managing and maintaining a building or facility. Their responsibilities may include ensuring safety and security, managing repairs and maintenance, coordinating services, and addressing tenant or occupant needs

Critical Level – refers to the quantity of school materials, equipment and other items that are low level in the inventory.

GS Department Head – refers to the person responsible for overseeing and managing a specific department within an organization. The department head typically has authority over departmental operations, budgeting, goal-setting, and personnel management.

Inventory Admin – refers to the person responsible for managing and controlling the inventory within an organization. Their tasks may include tracking stock levels, coordinating procurement and storage, conducting audits, optimizing inventory turnover, and implementing inventory management systems.

Kiosk – is a device used to provide information for borrowers. It is also used for data entry for the informations of the borrower who will request an item to the department.

Maintenance – refers to the process where the department head of General Services Department is adding, updating, and deleting informations on the system.

Procurement - The process of obtaining goods or services from external suppliers or vendors. Procurement involves activities such as supplier selection, negotiation, contract management, purchasing, and receiving.

Purchase Order – is the process after quotation. Once the item supplier is selected, the department personnel creates a purchase order specifying the items, quantities, agreed-upon prices, delivery dates, terms, and conditions.

Quotation – refers to the process where the department personnel evaluate potential suppliers based on criteria such as item price, quality, reliability, and delivery time.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter shows the related literature and related studies that are significant to the proposed project. It contains facts from books, thesis, journal, article, and electronic resources and further expands the idea of the study to develop a successful project.

2.1 REVIEW OF RELATED LITERATURE

FOREIGN LITERATURE

According to S S Islam, A H Pulungan and A Rochim (2019). Inventory refers to the supplies or materials that company uses for their production and sale. They also said that good inventory management is mandatory to maintain materials at optimal costs. And that companies need to implement better internal control, for example, by establishing policies, building employee capacity, preparing plans, and so forth, in order to increase inventory cost efficiency.

According to Tariq Hussain Sheakh (2018). Inventory management is a challenging problem area for most businesses and organizations. That is why corporations and organizations store inventory in warehouses to fulfill customer demand. The inventory management task is to find the number of items in the inventories to avoid overstocking. He also added that inventory is the raw supply of materials, finished goods called work-in-progress, and finished goods that an organization maintains to meet its operational needs and defined it as a good that is maintained by a business in anticipation of some future demand.

In the online article of WiSys (2022). Inventory management software give businesses and organizations more precise and efficient control over warehouse tasks like purchase order creation, space optimization, and stock-keeping. It also gives them access

to accurate, real-time information so you never run critically low on your most important stock. Automated Inventory management enhances the cycle counting and other routine inventory checking, relieving several common headaches associated with manually verifying their stock levels.

According to Iluore et al. (2020). Asset monitoring is a physical asset management function that determines the present status, location and usage of the items and equipment of a company asset that need to monitored and tracked. In the manual process of doing things where companies and organizations are hiring personnels to monitor each asset and make a paper-based reports on its status. Real-time monitoring of assets is currently replacing the manual method of monitoring and tracking of assets since it has been proven to be unproductive to fulfull the goal of a company asset management unlike in the past.

According to Robert Baran (2018). Barcoding Technology has a direct impact on Inventory Management. Since inventory management is a hot topic in today's world of fast-paced and continuously evolving technologies, Barcode scanning has totally increased the efficiency of businesses and organizations by making it easier to identify items in the inventory, track them, and manage the ordering process. Additionally, barcoding technology helps reduce the amount of time and resources used in inventory management by streamlining the process. It also helps reduce human errors and improve accuracy.

LOCAL LITERATURE

According to Vitez, O. (2018). Inventory is the accounting for the total goods, services, and materials in a factory, store, or other business. Inventory could be done manually, but the added requirement for businesses to track, monitor, and effectively manage the inventory of stocks or services offered by the company is necessary in these modern times. With the use of modern technology, such as computers, businesses can very

well reduce the cost of their operations, produce real-time reports, and create a secure environment to protect the customer and the company's information.

According to Bautista & Young (2022). Inventory management is one of the essential processes in procurement and logistics management. Inventory is one of the most important assets of any company, especially those on the production line. Inventory issues, such as out-of-stocks, disrupt a company's business processes. An out-of-stock situation occurs when a company is unable to keep up with customer demand. Therefore, the task of effective inventory management is to ensure that inventory levels are sufficient to meet demand without overstocking.

According to Bacudbacud, S. et al. (2020). An inventory management system typically includes software that tracks inventory levels, monitors stock movements, and records information about the product, such as price, expiration dates, brands, categories, and supplier information. Also, it is a system that oversees the monitoring and maintenance of their stocked, whether those products are assets, raw materials, or supplies of company that are ready to be sent and lent to consumers. The inventory management system reduces user errors and manages the miscalculation of all supplies by handling the item record and creating a a quick way to track and sort, add equipment and items to ensure that data is accurate error-free and up-to-date.

2.2 REVIEW OF RELATED STUDIES

FOREIGN STUDIES

According to the study by Esha Saha and Pradip Kumar Ray (2019).They discusses the significance of inventory management systems in healthcare, especially in hospital systems. Inventory management systems play a crucial role in managing and controlling a large and diverse range of items that are necessary for patient care, monitoring, and control. While cost-based models are often used in inventory management systems, it is important

to prioritize patient service levels. In a constantly fluctuating environment, it is challenging to ensure the availability of high-quality healthcare items. Maintaining an appropriate inventory level is crucial to ensure confidence in the healthcare system. Inadequate inventory levels can lead to stock-outs and a loss of faith in the system, while a high level of inventory can result in excessive investment and capital unavailability for other purposes. Predicting demand accurately in healthcare systems is also challenging due to uncertainties and randomness such as varying patient conditions, physician prescriptions, and individual patient responses to treatments. The article emphasizes the need for healthcare organizations to focus on optimizing inventory management systems to ensure optimal patient care and efficient use of resources.

The study of S. G. Srivatsa et al. (2021) presents a system that uses sensor-based IoT technology to remotely monitor and track perishable goods. The system aims to provide early warning alerts and notifications in critical conditions, ensuring end-to-end visibility and accountability across the entire product value chain. With perishable goods often required to travel long distances, our proposed system offers real-time monitoring to prevent inventory system damage and minimize losses. According to our research, there are 7645 cold storage facilities in India, with around 68% used for vegetable storage.

The study Tejesh and Neeraja (2018) discussed the importance of the warehouse inventory management system using IoT and open-source framework. The warehouse is the section where products or goods are stored and its prime goal is to control the flow of these items. Effective management of products is crucial as it can impact time and cost. With the globalization of industries, warehouse inventory management systems have gained significant importance due to the profits they can generate. Warehousing involves storing products on a large scale in a precise manner and ensuring their availability when needed. Warehouses are needed to store different types of products to maintain seasonal production, seasonal demand, quick supply, continuous production, and price stabilization. A computerized warehousing system is essential for every warehouse, as it requires less

effort, provides more efficiency and stability when compared to a manual system. Warehouses consist of multiple zones, which are also referred to as stockrooms. Depending on the demand for products, more stockrooms may be required. Automating warehouses is necessary to avoid human errors that may impact warehouse utilization.

Masandig et al. (2022) discussed the development of a digital stock record keeping system called E-Card Stock Web Based Using Barcode for Fast Kitchen Concept. The inventory system is used by traders in business management to obtain current quantity stock of raw materials or products. Retailers in the retail sector, particularly those offering basic goods such as food and beverage products, require efficient and effective product stock management to prevent disruptions in the food chain or delays due to inaccurate stock items. However, some retailers still manage their stock manually, counting and recording it in a book or stock card, which can lead to various errors, such as information not being updated, unclear writing or loss of record books. To address this issue, this project utilized barcode technology to develop a stock information system that can be stored in a database. Barcode technology is a scanning technology that can be translated into a code understood by the software system. The implementation of E-Card Stock Web Based Using Barcode for Fast Kitchen Concept aims to provide a digital stock record keeping system for efficient and effective management of stock in the retail sector.

Van Groesen and Pauwels (2022) discuss the challenges faced by the construction industry in terms of trust, information sharing, and process automation. To increase transparency and trust, they suggest addressing physical asset tracking (PAT) and digital asset management (DAM). They propose that various technologies and workflows could aid this purpose, such as geospatial tracking technologies and related asset management systems. The authors note that these technologies are already widely applied outside the construction industry to replace manual asset tracking, and they are increasingly being used in construction as well. Additionally, they point out that the combination of building information modeling (BIM) models with real-time data could enhance current practice.

Emerging technologies such as blockchain and smart contracts are also identified as promising contributors to transparency and trust. Although there are still challenges to the joint application of these technologies, the authors suggest that the construction industry is ready for a socio-technical shift towards their integration. Van Groesen and Pauwels aim to address the challenges of trust and transparency by combining and testing these technologies in an industry use case. Specifically, they aim to semi-automate asset and compliance tracking activities for large prefabricated assets throughout the supply chain.

LOCAL STUDIES

According to A. Diaz, T. et al. (2021). Development of a Hotel Inventory System through Agile Methodology. The study presents Hotel ABC's daily inventory experience, their operational problems, and the proposed solution by the researchers using an Agile methodology. The absence of an automated inventory system at Hotel ABC resulted in manual storage of all data and records, causing bottlenecks in their operations. The proposed hotel inventory system aims to eliminate these problems, reduce data inaccuracy and loss, and provide a complete list of products and their respective quantities to avoid overstocking and understocking. The system also monitors returnable and consumable products, their locations, and secures data accessibility for authorized personnel (admin, senior manager, accounting, housekeeping, and purchasing departments).

Equipment Reservation and Inventory System (2020) an Equipment Reservation and Inventory System was developed to establish a technological approach to equipment reservation in local institutions. The system aimed to empower institution personnel and improve task efficiency. The study aimed to develop an automated system that follows user commands with accuracy and consistency, beneficial to the company and personnel. This concept emphasized the collaboration between people and computers to achieve specific goals. The rapid evolution of technology has led to breakthrough software development, which institutions have adapted to improve work speed. The system's significance has

increased from being used solely in factories and industries to becoming an indispensable tool in institutions.

Inn Management and Reservation System (2022) The Middle Town Inn currently uses a manual reservation system that is time-consuming for both customers and staff. Customers need to fill up forms with their personal information, which are then manually stored in a wooden cabinet with other customer files. The calculation and inventory are also done manually, and changing customer information requires the receptionist to search through physical files. Insufficient storage and lack of security of the files result in lost customer records. To improve the business transactions, the proponents propose a Management and Reservation System that is more secure, efficient, and accurate than the manual process. The proposed system allows customers to view and reserve their desired room online, with descriptions such as room number, rate, and location available upon reservation. The system can add, update, delete, and print records, and customers receive a notification of their successful reservation via email. The system can also retrieve and restore data and provide different forms for various event reservations. The database keeps track of customer records and transactions.

Nemendez (2022) presents a real-time implementation of an inventory management system for on-site businesses such as restaurants and grocery stores. The proposed system aims to efficiently monitor the flow of units into and out of the inventory to prevent overstocking or stock shortage that could negatively impact the company's operation. The system is particularly important for businesses that receive a high volume of orders and require proper maintenance of materials. Additionally, the system records the time taken to process an order and alerts the organization when the inventory level is low. The system focuses on three aspects of inventory management, namely control of inventory transfer, understanding of lead time, and proper ordering to prevent future failures. The study emphasizes the importance of having a solid understanding of lead time to determine when to place an order and how many units are necessary to ensure smooth production.

Daigo et al. (2019) study present a Lan-based inventory with barcode scanner for Coromano Marketing. The study said that the manual system is now considered obsolete after the birth of computerized system. It becomes more attractive to the clients considering it can save time and hassle free. Moreover, the inventory system promotes effective inventory control which ensures stocking the in demand and correct items in correct quantities. Having an advance system on LAN-based and inventories makes company more productive, efficient, and convenient to the company. The proposed system is meant to help people show to customer more relevant items, hoping to expedite and increase the LAN-based inventory systems, management can easily make consistent, reliable, and timely decisions.

The study of Delos Santos et al. (2019) discussed a LAN-based Point of Sale with inventory system using barcode scanner it is a system that helps the company or a business, a point of sale or point of purchase where a transaction is carried out once customer picks the products for buying, inventory and track sales activity. It is one of the most systems that are used of any business. It helps to improve the process and to fasten the transactions with their customer.

2.3 SYNTHESIS OF THE REVIEWED RELATED LITERATURE AND STUDIES

The proponents believe that all references and studies mentioned in this study are similar to the current study. The proponents also believe that the same methods and techniques used in the studies mentioned in the study can be applied to the current study. Additionally, the proponents believe that the results of the current study can be compared to the results of the references and studies mentioned in the study to provide a more comprehensive understanding of the topic.

The related literature, both foreign and local, all stated that inventory management is essential to being competitive in this day of new-age technology. Inventory management involves managing the goods and products held by a company. It involves tracking the quantity, cost, and location of items, as well as ordering, storing, and managing the flow of goods. The references used also stated that Barcoding technology plays a significant role in inventory and asset tracking management. Because it involves the use of barcodes, which are machine-readable labels containing information about a particular item or asset. It makes tracking and identification of items more accurate by reducing human error and time-consuming manual processes. The barcode technology also helps not only on inventories of the companies but also on the side of managing the company assets by providing accurate information about the whereabouts of an item or asset, which is essential for effective inventory control.

The related studies, both foreign and local, give examples of systems that are similar to the proposed projects of the proponents. Inventory management is not only helping businesses but also organizations that are monitoring their assets. In the study of Esha Saha and Pradip Kumar Ray (2019). They discuss the significance of inventory management systems in healthcare, especially in hospital systems. Inventory management systems play a crucial role in managing and controlling a large and diverse range of items that are necessary for patient care, monitoring, and control. Another study named

"Smart Cold Storage and Inventory Monitoring System" by S. G. Srivatsa, K. R. M. Bharadwaj, S. L. Alamuri, M. M. C. Shanif, and H. S. Shreenidhi (2021) presents a system that uses sensor-based IoT technology to remotely monitor and track perishable goods. Their system aims to provide early warning alerts and notifications in critical conditions, ensuring end-to-end visibility and accountability across the entire product value chain. While the study of Masandig et al. (2022) discussed the possible benefit of the development of their digital stock record-keeping system called E-Card Stock Web-Based Using Barcode for Fast Kitchen Concept. Their system utilized barcode technology to develop a stock information system that could be stored in a database, providing them with a digital stock record-keeping system for efficient and effective stock management in the retail sector. All these studies provide an insight into the benefits and importance of the development of inventory management systems with barcode technology, and the proposed projects of the proponents can be developed based on the ideas presented in these studies.

The proponents can use the same technology used in these studies and develop their own system that is tailored to the specific needs of their business. In addition, the proponents can also use the findings of the studies to identify the best practices for developing their own system.

CHAPTER III

METHODOLOGY AND RESEARCH DESIGN

This chapter discusses the research design of the proposed system. It also contains, the methodology used, processes, information, and diagrams that represent the proposed system.

3.1 RESEARCH METHOD AND DESIGN

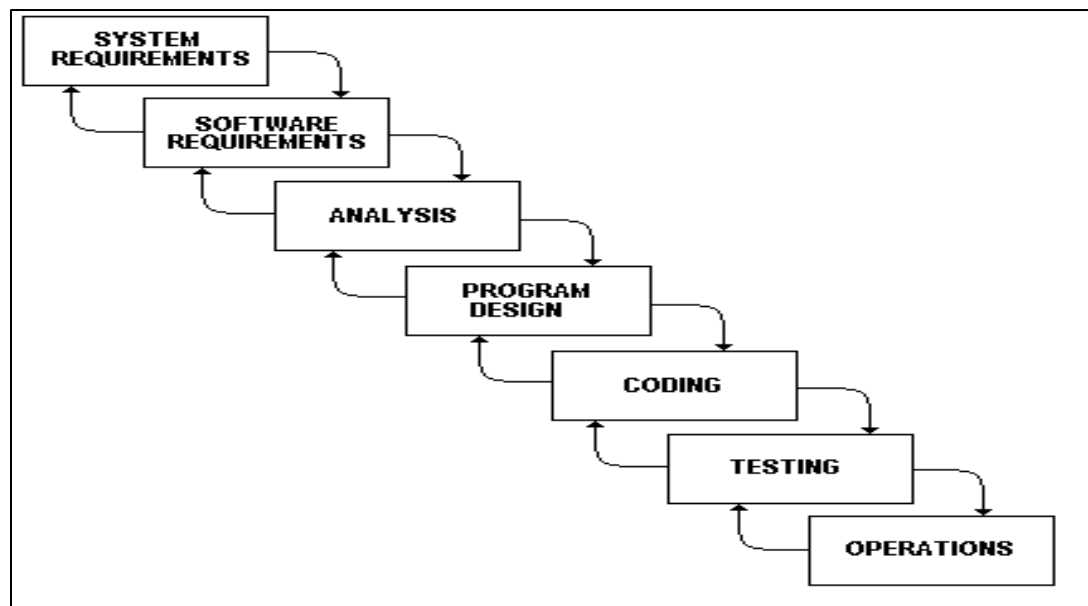


Figure 1: Research Method and Design (Modified Waterfall Model)

Figure 1 shows the method used in this study is the Modified Waterfall Model. The Modified Water Fall method is a derivative of the traditional water fall model but with some minor variations relative to iterations between certain stages. The modified waterfall provides an orderly sequence of development steps with some flexible iterative stages to facilitate the adequacy of documentation and design reviews to ensure the quality, reliability, and maintainability of the developed custom software.

Following the design phase, the proponents needed to know what the system and software requirements would be, followed by an analysis of whether they were valid or not. Once the requirements analysis is complete, the next phase is indeed the design phase of the system's user interface. This phase focuses on designing the visual and interactive elements of the system that users will interact with. The goal is to create a user interface that is intuitive, efficient, and visually appealing, while meeting the requirements and objectives of the project.

After the program design phase is done, the next phase the proponents will do is the coding of the system. In this phase, the proponents translate the system design into actual code and develop the software components that make up the system. The coding phase is a critical stage where the design concepts are transformed into working software. The output of this phase is a functional system that can be further refined and tested in the subsequent phases of the software development life cycle.

After the coding and testing phase, the system is ready to be deployed and used for operations. It is important to note that the operations phase is not the final step in the software development life cycle. Once the system is live, ongoing maintenance, updates, and enhancements are typically required to ensure its continued functionality, security, and alignment with evolving departments requirements.

3.2 CONCEPTUAL FRAMEWORK

The conceptual framework contains the methods and requirements in developing a Inventory Management and Asset Tracking System using barcode for Global Reciprocal Colleges General Services Department.

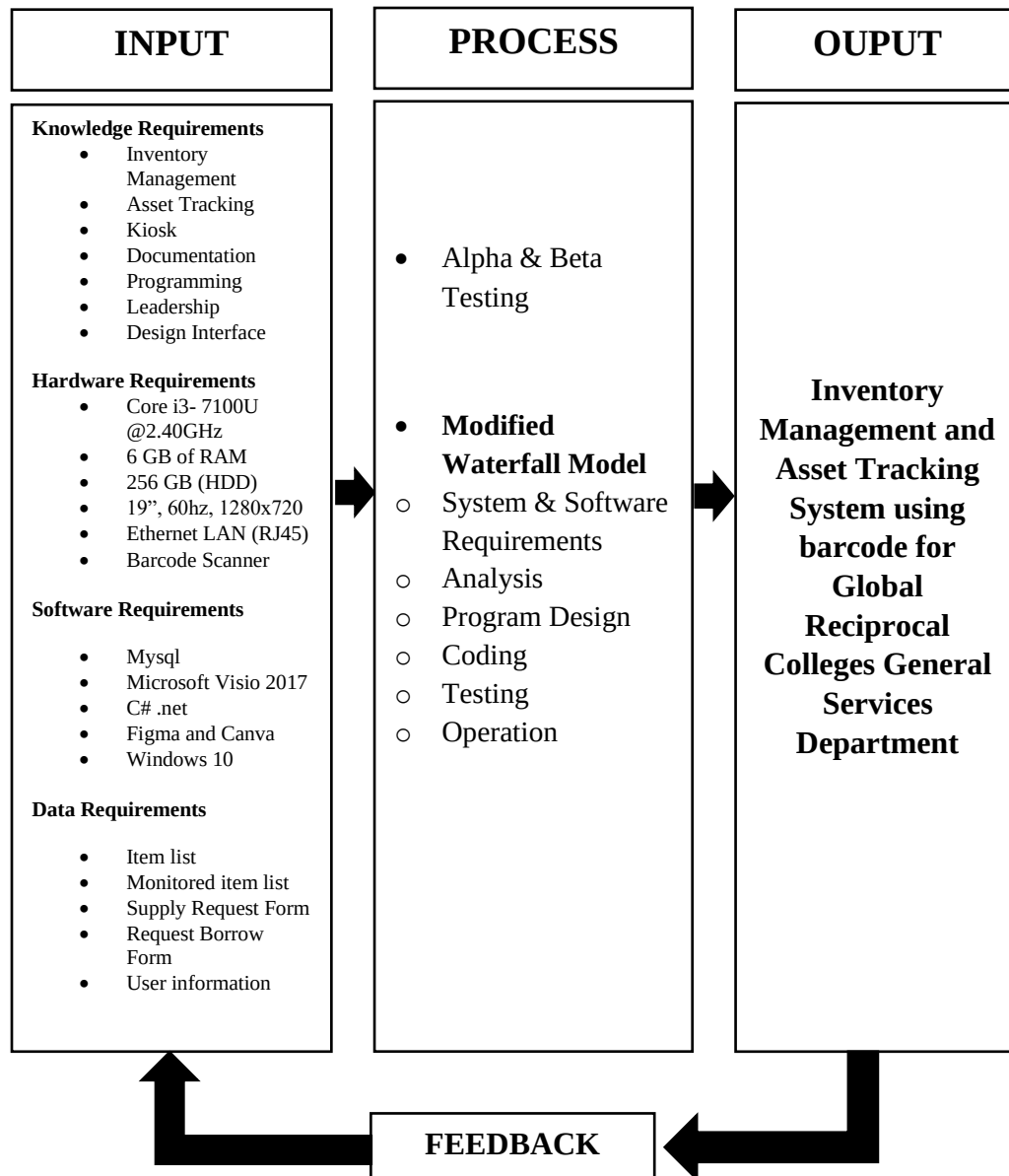


Figure 2: Conceptual Framework of the Study

Figure 2 shows how the system will be constructed, the study's conceptual framework is separated into parts; the input, process, and output. In input phase, it consists of Knowledge, Hardware, and Software requirements which are needed for the development of Inventory Management and Asset Tracking System using barcode for Global Reciprocal Colleges General Services Department.

In the Input phase, Knowledge Requirements identifies the source of information upon developing the system. These are the primary ideas proponents must know on what is a inventory management and how to track the assets of a organization without using a tracking devices, how data is shared through Local Area Network (LAN), and how the system can provide the status of procurement. Knowledge Requirement also includes the skills of the proponents to build the proposed system and create this project. In Hardware Requirements are the materials needed to run the system on the users' devices. While Software Requirements are the software used to create the system. And Data Requirements are the informations provided by the department to be incoded in the system.

The process phase contains the software model to used. The proponents used the Modified Waterfall Model in creating the system. The modified waterfall provides an orderly sequence of development steps and if errors are detected in any of the phase the proponents can go back to previous phases to correct the errors occurred. The proponents will do Alpha and Beta testing to test the proposed system. The proponents will do Alpha testing to identify bugs before giving the system to the General Services Department. After the Alpha testing is completed, the proponents will typically move on to the next phase, which is Beta testing. Beta testing is conducted to further evaluate the system's performance, stability, and compatibility in a real-world environment. Unlike alpha testing, which is usually done in a controlled setting by the development team, beta testing involves a wider range of users or selected participants who are not directly involved in the system's development. Once the beta testing phase is successfully completed and the system has been refined based on user feedback, it can be considered ready for deployment to the General Services Department or other end-users.

3.3 HARDWARE AND SOFTWARE REQUIREMENTS

Table 1: Hardware Requirements

HARDWARE COMPONENTS	MINIMUM REQUIREMENTS	RECOMMENDED REQUIREMENTS
Processor	Core i3- 7100U @2.40GHz	Core i5-7200U @ 2.70GHz
Memory	6 GB of RAM	8 GB of RAM
Hard Drive	256 Gigabytes (HDD)	500 Gigabytes (HDD)
Monitor	19" wide, 60hz, 1280x720	22", 60 to 75 hz, 1280x720
Wired Networking	Ethernet LAN (RJ45) & Router	
Barcode Scanner	1D USB Wired Barcode Scanner	

Table 1 shows the minimum and recommended requirements to run the proposed system. The system would run on a Core i3-7100U @ 2.40 GHz processor with 6 GB of RAM and 256 Gigabytes of storage, but would run better on a Core i5-7200U @ 2.70 GHz processor with 8 GB of RAM and 500 Gigabytes to a terabyte of storage. The monitor of the user will also affect the interface of the system, so the monitor of the user should be atleast 19 inches wide with 60 Hz and have a resolution of 1280 x 720 . The system will also need an RJ45 cable and a router to enable communication between devices within the same network. And ofcourse the barcode scanner for checking out the items.

Table 2: Software Requirements

SOFTWARE COMPONENTS	USED SOFTWARE
Database designer	mysql, Microsoft Visio 2017
Software Language	C# .net
User Interface Designer	Figma, Canva
Operating System	Windows 11

Table 2 shows the software components used to develop the proposed system by the proponents. The proponents used mysql and Microsoft Visio 2017 for implementing diagrams for the database of the system. C#.net is the programming language used to develop the proposed system. The proponents used both Figma and Canva for designing the User Interface of the proposed system. Figma was used to create modern looking layout design with appropriate color scheme and typography. Also, Canva was used to create minimal yet attractive elements such as icons, buttons, and illustrations. Additionally, both Figma and Canva were also used to create attractive visual elements such as progress bars, diagrams, and grids. These visual editor were used to help engage with the users and make the system more user friendly. And the system is created on a Windows 11 operating system.

3.4 CONTEXT DIAGRAM

EXISTING

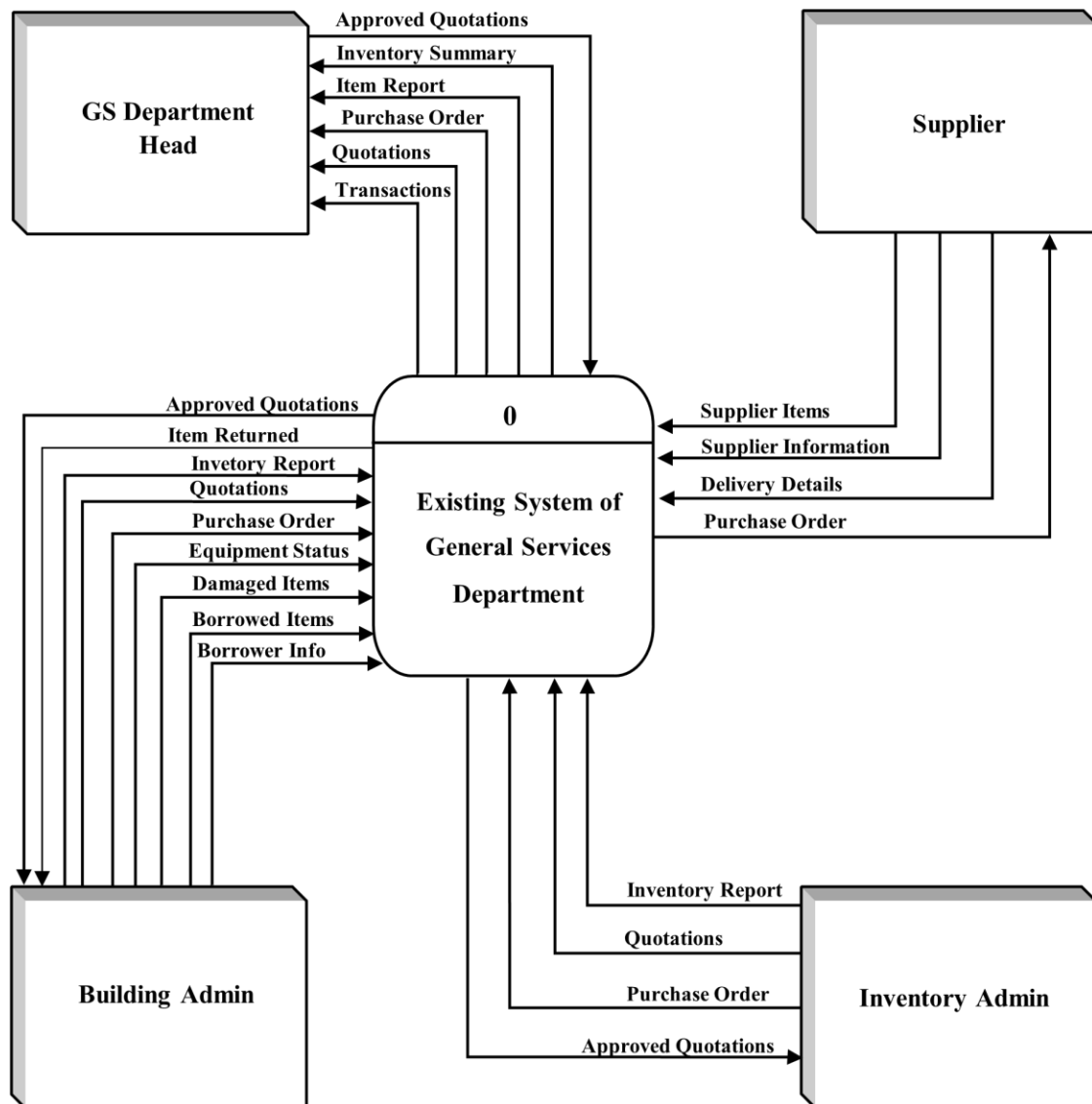


Figure 3: Context Diagram of General Services Department Existing System

Figure 3 represents the diagram of the existing system of General Services Department. All the transactions and reports in the department are given to the department head. Building Admin is responsible for the tools and equipment of the school. He is also responsible for writing a report of the equipment's status to know if it is damaged, borrowed, or not. While the inventory administrator is the one who is responsible for the consumable items of the department and makes sure that the items in the inventory are organized on their respective shelves. Both building and inventory administrators are required to hand in their quotations to the department head to get approval. And after quotations are done, they can now proceed to their purchase order to be signed by the higher-ups.

PROPOSED

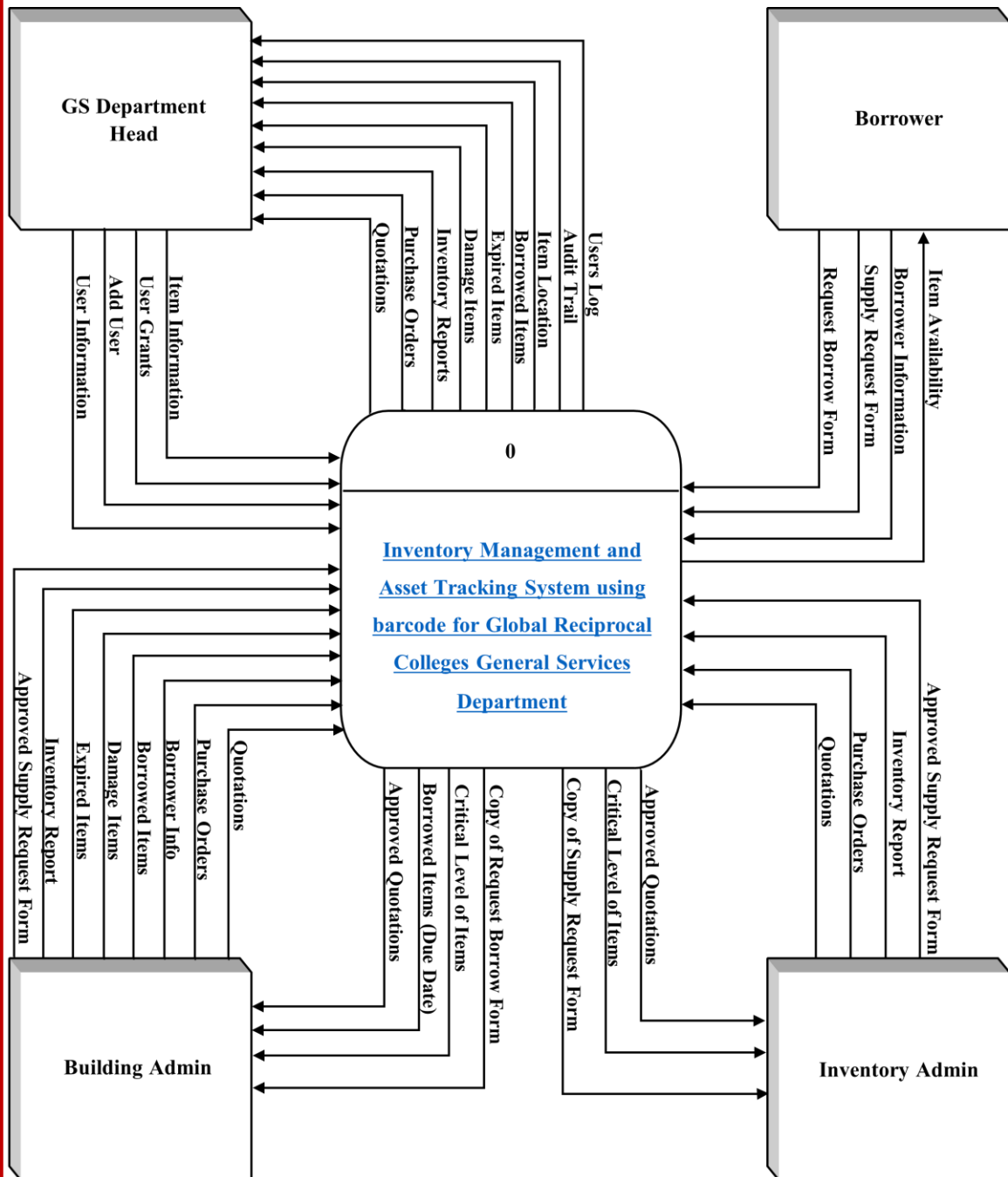


Figure 4: Context Diagram of General Services Department Proposed System

Figure 4 represents the proposed diagram of the General Services Department. The department head plays a central role in the proposed system. They are responsible for granting access and accounts to the inventory and building administrators. Additionally, they have the authority to monitor all transactions and user logins within the system. Inventory and building admins are responsible for managing the inventory and building-related activities. They perform tasks such as maintaining records, updating inventory levels, managing building resources, etc. The department head can monitor all transactions within the proposed system. This monitoring likely includes tracking changes made to inventory, building resources, and any other relevant transactions performed by the admins. The borrower is the only entity who is not part of the department's staff. The role of the borrower is to put their information and their desired item to request and borrowed from the department through the device just outside the GS department office.

3.5 DATA FLOW DIAGRAM

EXISTING

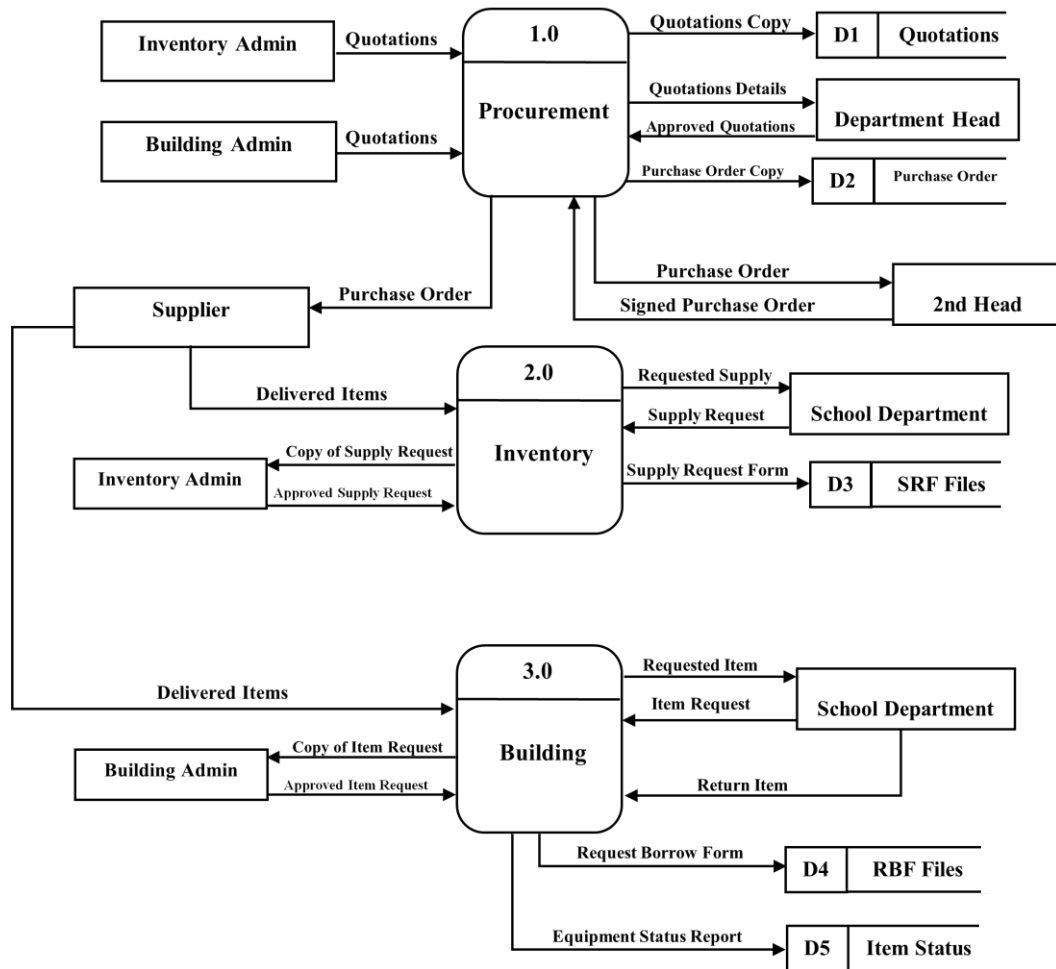


Figure 5: Data Flow Diagram of Existing System

Figure 5 represents the existing data flow of the General Services Department. This diagram shows the existing processes of the department. The first process is where quotations and purchase orders are made. Both inventory and building admin are expected to hand over quotations for items that need to be ordered. After the department head approves the quotations, a copy of the quotation will be handed to the second head of the department to sign. All of their quotations and purchase orders are stored on a spreadsheet in case their higher-ups need to see those transactions. A copy of the signed purchase order will be handed to the supplier. After the supplier receives a copy, they will now process their order and deliver it to them.

After the delivery, the inventory and building admins take over the process. They are responsible for receiving and verifying the delivered items. The inventory admin, upon verification of the delivered items, updates the inventory records to reflect the new stock. This involve updating quantities, item details, and other relevant information in the inventory. The building admin, on the other hand, replenishes the building resources using the delivered items. This include restocking supplies, equipment, or any other resources required for building maintenance or operations.

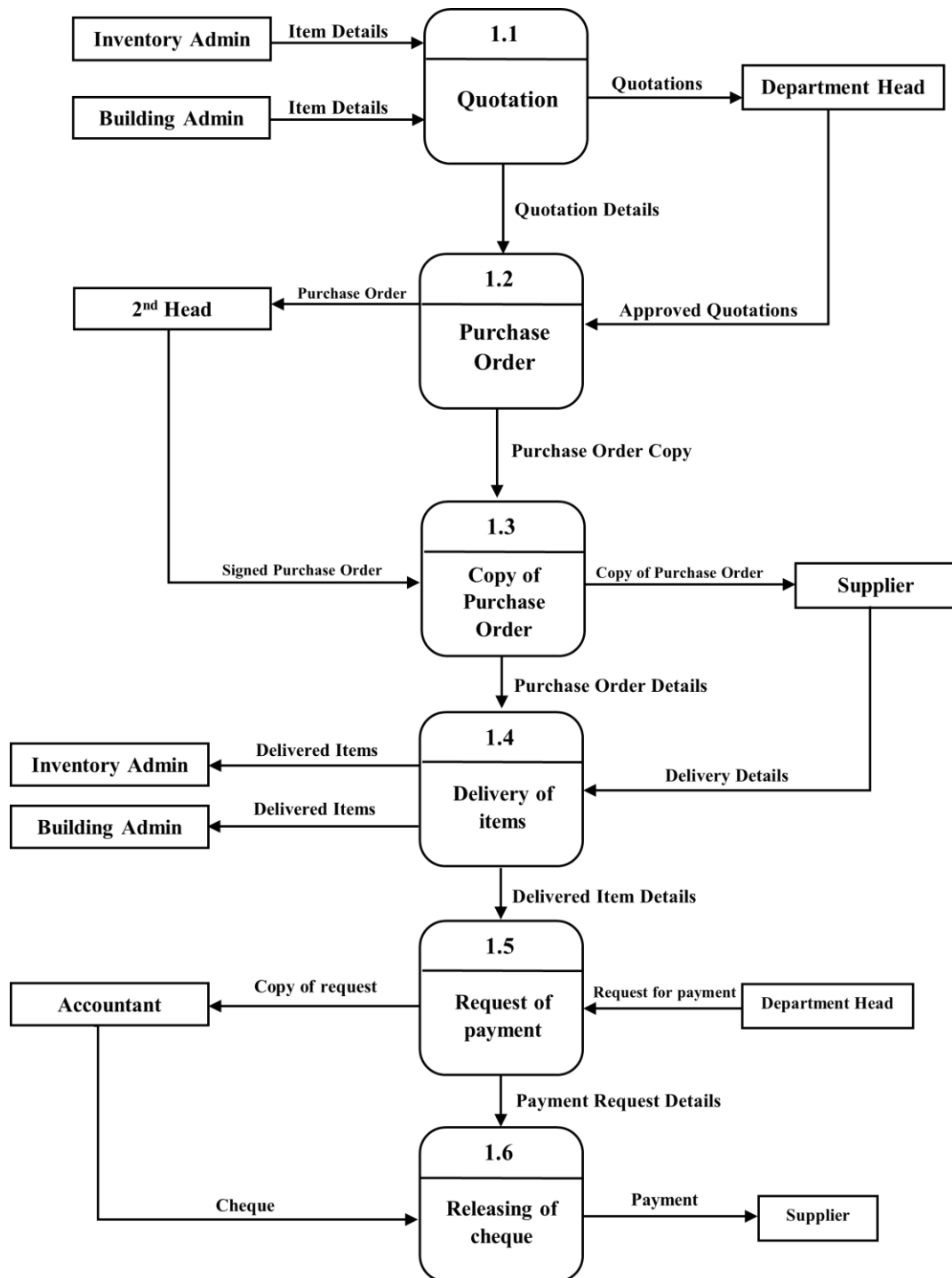


Figure 6: Data Flow Diagram of Process 1 (Existing)

Figure 6 represents the data flow diagram of process 1 of the existing system of General Services Department.

This Diagram show the data flow under the existing diagram 1 of process 1. The process will start with the inventory and building admin handing over the item details on what they need to order. The department head will now check whether the quotation is approved or denied. The approved quotations will now be a purchase order and are ready to be handed to the second head of the department for signature. After they sign the purchase order, a copy of the purchase order will be handed to the supplier for them to process and deliver the items. Once the items are delivered, the department head will hand a request for payment to the accountant. As soon as the request for payment is approved, the department head will give it to the inventory and building admin to pay the items and hand it to the supplier.

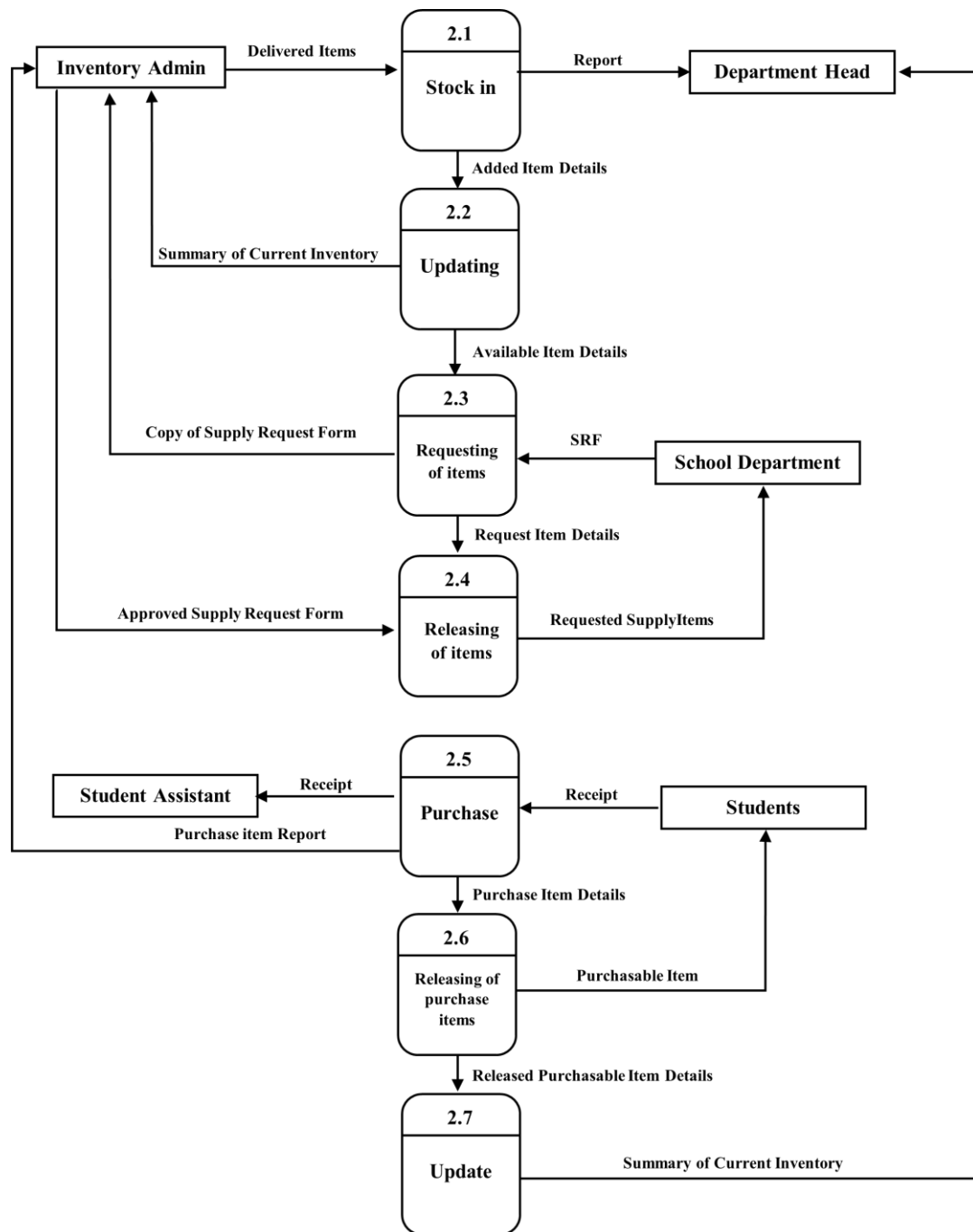


Figure 7: Data Flow Diagram Process 2 (Existing)

Figure 7 represents the data flow diagram of process 2 of the existing system of General Services Department.

Once the inventory admin receives the delivered items and performs a thorough check to ensure the accuracy and quality of the items. The inventory admin can now update the inventory records to reflect the new stock. This includes recording the quantities, item details, and other relevant information in the inventory. With the inventory restocked, the General Services Department can now facilitate the process of fulfilling requests from students and school departments for supplies. Students and school departments can now submit requests to the General Services Department specifying the supplies they require for their respective needs within the school. The inventory admin evaluates the supply requests to determine if the requested items are available in the inventory and if the requested quantities are reasonable. They may also consider any priority or urgency assigned to the requests. Once the evaluation is complete and the requested items are available, the inventory admin proceeds to distribute the supplies to the requesting students or school departments. This can involve recording the distribution details and updating the inventory records accordingly.

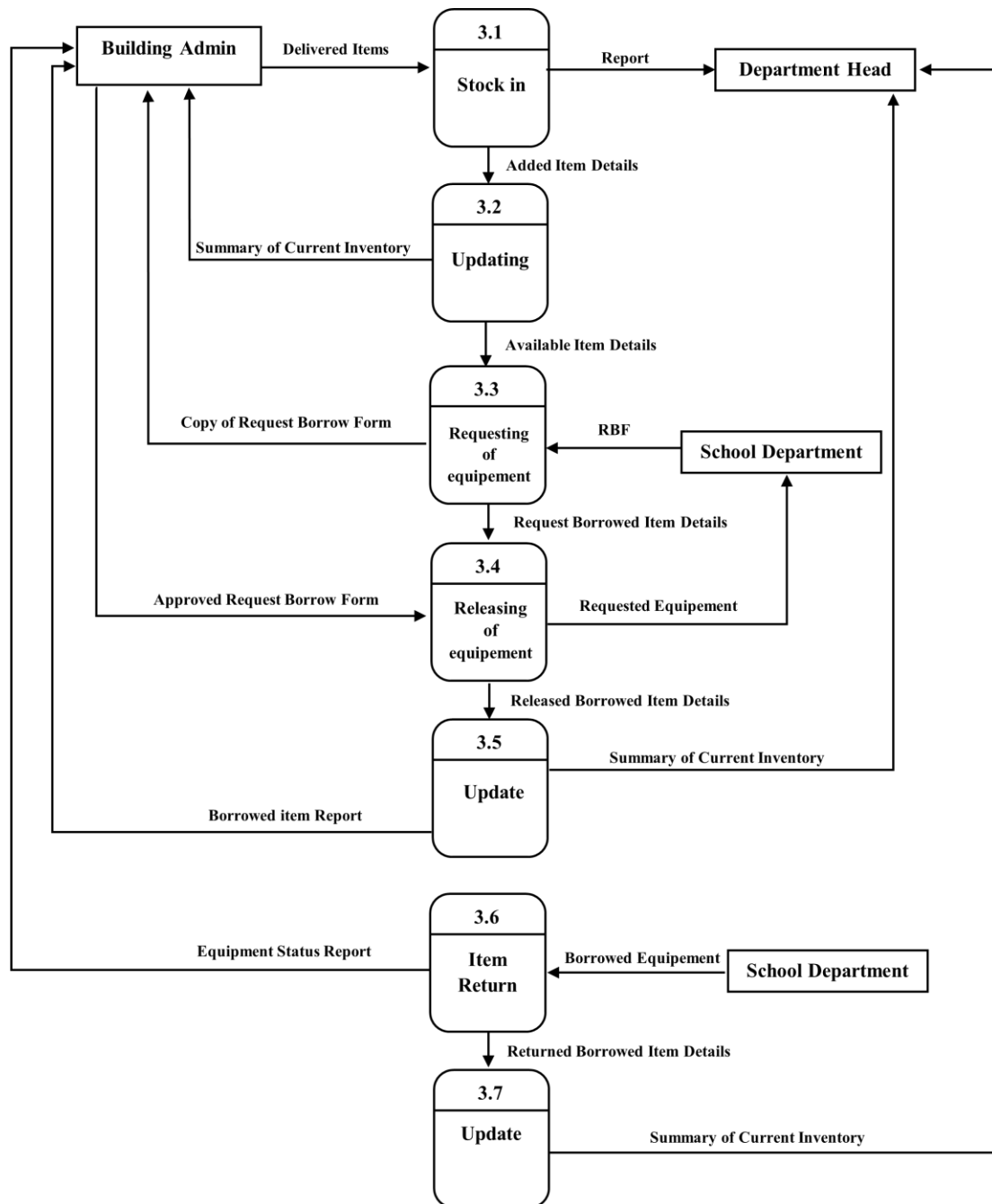


Figure 8: Data Flow Diagram of Process 3 (Existing)

Figure 8 represents the data flow diagram of process 3 of the existing system of General Services Department.

After the building admin received the items from the supplier. he can now process the requested items from them.

Once the building admin receives the delivered items and performs any necessary checks or inspections to ensure the quality and correctness of the items. The building admin can now update the inventory records to reflect the new stock. This includes recording the quantities, item details, and other relevant information in the inventory. After the inventory for building resources is restocked, the General Services Department can now facilitate the process of fulfilling requests from students and school departments to borrow items for use inside the school. Students and school departments can now submit their requests to the General Services Department, specifying the items they need to borrow for their respective purposes within the school. The building admin will evaluates the borrowing requests to determine if the requested items are available in the inventory, and if they are suitable for borrowing. They may also consider any priority or restrictions associated with the requests. Once the evaluation is complete and the requested items are available, the building admin allocates the items to the requesting students or school departments. This may involve tracking the item allocation and the expected return dates. After the students or school departments have finished using the borrowed items, they return them to the General Services Department. The building admin receives the returned items, performs necessary inspections, and updates the inventory records accordingly.

PROPOSED

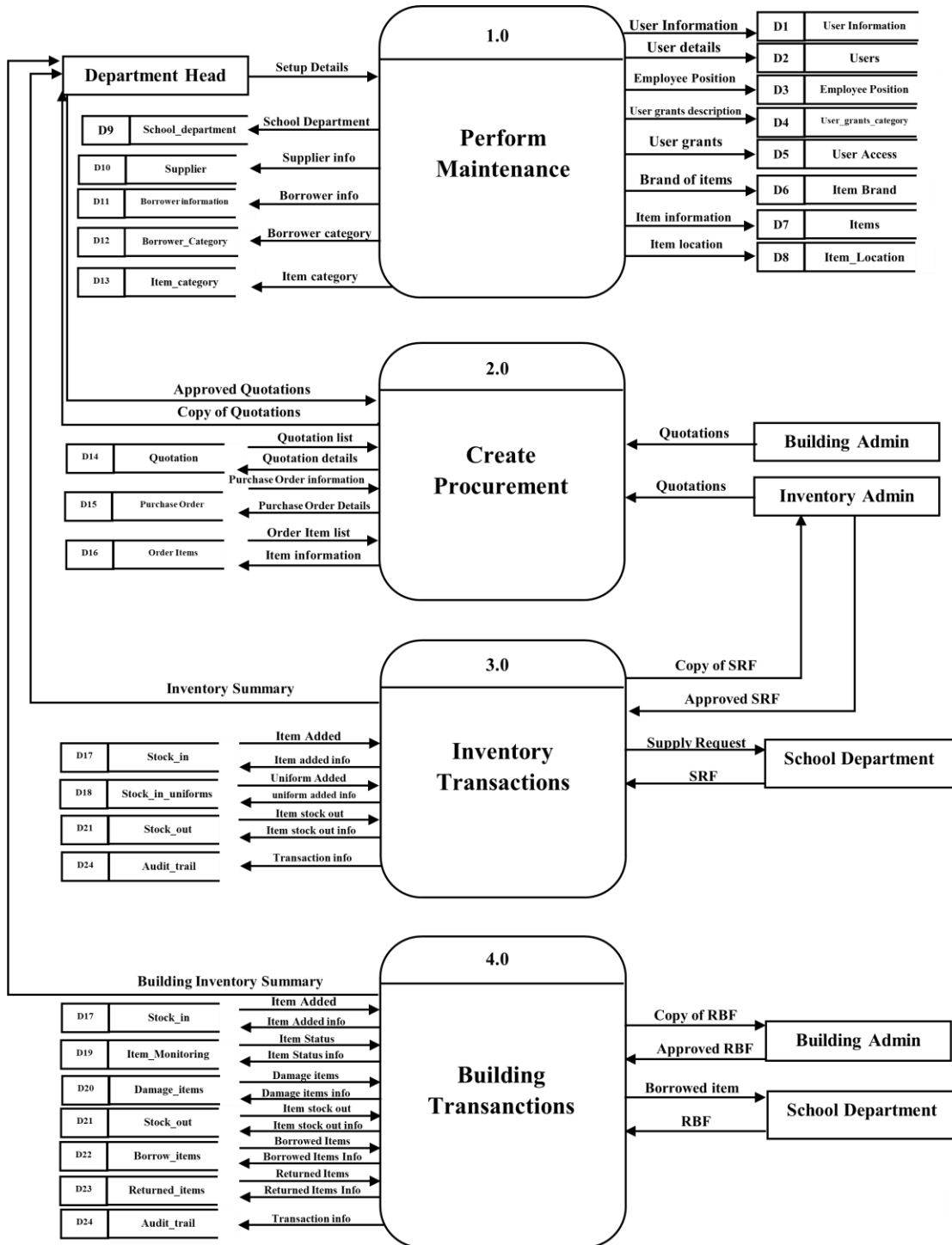


Figure 9: Data Flow Diagram of Proposed System

Figure 9 represents the data flow diagram of the proposed system for General Services Department.

The process will start at the department head after they finished the setup details for the maintenance of the system. It implies that the department head plays a crucial role in initiating the process and ensuring that the system is set up properly for ongoing transactions. This setup phase involves configuring the necessary components, establishing maintenance protocols, and ensuring that the system is ready to be utilized by the department. After the maintenance is done where the department head finishes the set up, the next process is the procurement, this process will only be available when the inventory and building admin handed a copy of their respective quotations. Once the quotation details are approved the two admins can now print their respective quotations and are ready to hand a copy to the second head where it needs to be sign in able to start the delivering of the items.

The next process is when the items are delivered, inventory and building admin can now start their respective task in the system. The inventory admin is responsible for distributing the supplies to the school departments as needed. This involves keeping track of the inventory levels, monitoring requests from various departments, and ensuring that the necessary supplies are provided promptly. This includes monitoring stock levels, replenishing supplies when necessary, and updating the inventory records accordingly. While the building admin's role is to lend items to the school department when they require them. This could include equipment, tools, or other resources necessary for the department's activities. The building admin is also responsible for maintaining a record of the items that are lent out. They will track the borrowing and returning of items, ensuring that all transactions are properly recorded and accounted for.

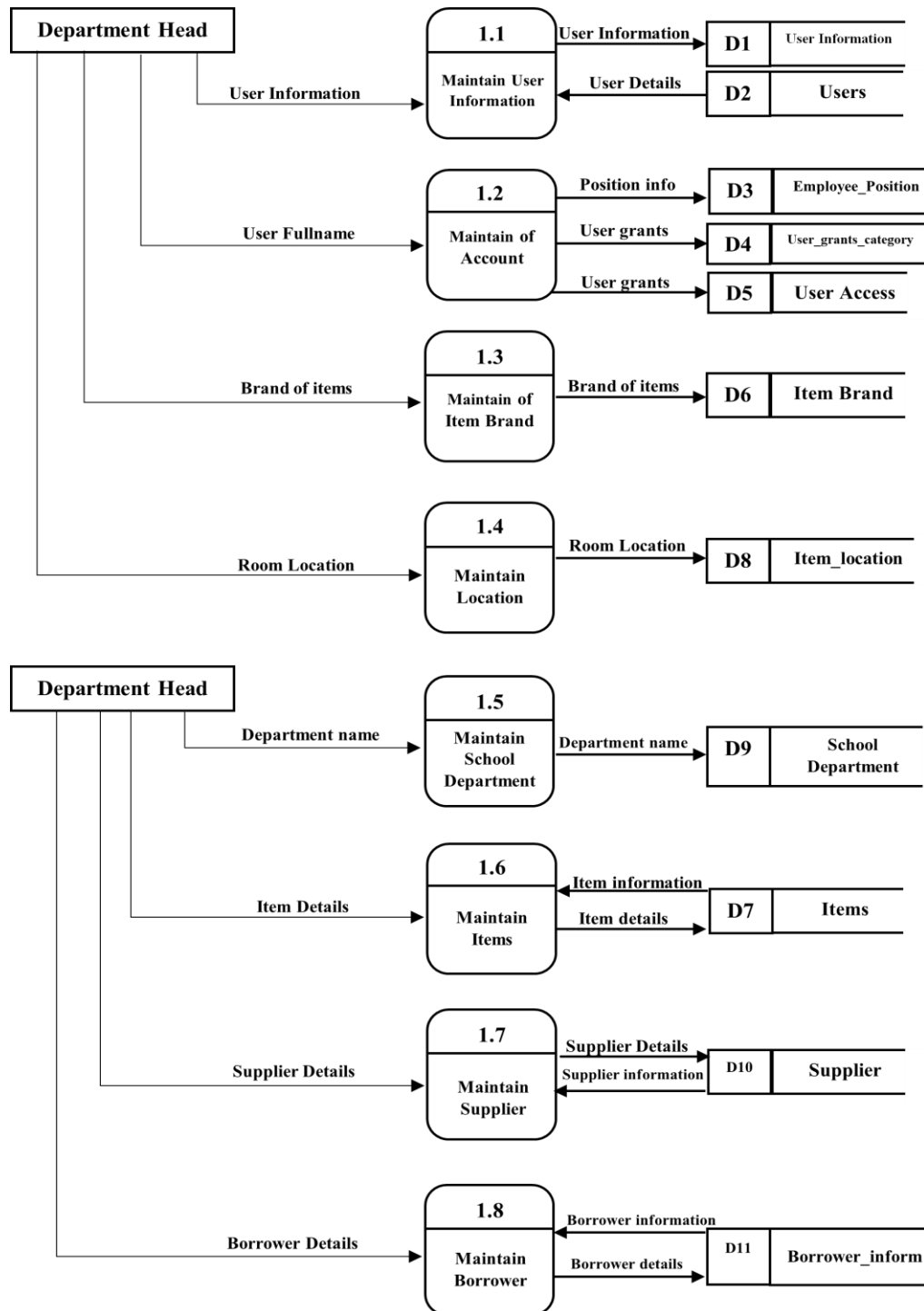


Figure 10: Diagram 1 of Process 1 (Proposed)

Figure 10 represents the data flow diagram process 1 of the proposed system of General Services Department. This diagram shows the maintenance module of the system. This module contains sub modules where authorized users like administrator can add information such as user, employee position, user access, item brand, item location, school department, items, supplier, and borrower information. Likewise, all the information that will be saved or processed are validated upon clicking the save button or update button. This implies that any part of the system that is adjustable or changeable is included in the maintenance scheme. This indicates that the maintenance module allows for easy customization and adaptation of the system as per the evolving needs of the General Services Department.

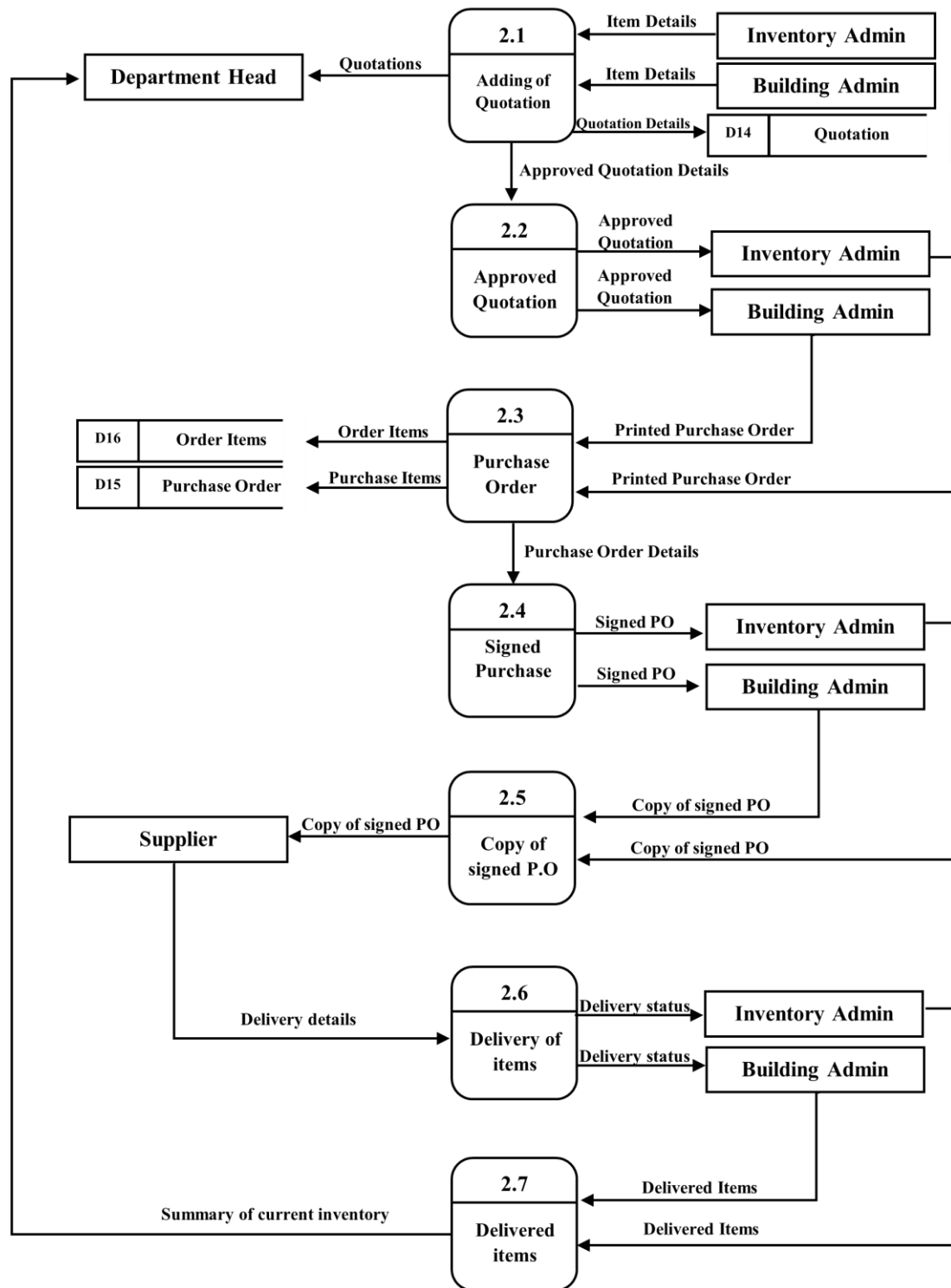


Figure 11: Diagram 1 of Process 2 (Proposed)

Figure 11 represents the data flow diagram of process 2 of the proposed system of General Services Department. This diagram shows the quotation process of the proposed system. The process will start with the inventory and building admin handing over the item details on what they need to order. The department head will now check whether the quotation is approved or denied. The approved quotations will now be a purchase order and are ready to be handed to the second head of the department for signature. After they sign the purchase order, a copy of the purchase order will be handed to the supplier for them to process and deliver the items. The system will provide the status of the delivered items. Once the items are delivered the inventory and building admin will proceed to check those delivered items. After thorough checking the two admins can now update their respective inventories. All transactions made in the system will be stored on a database where the department head can see it.

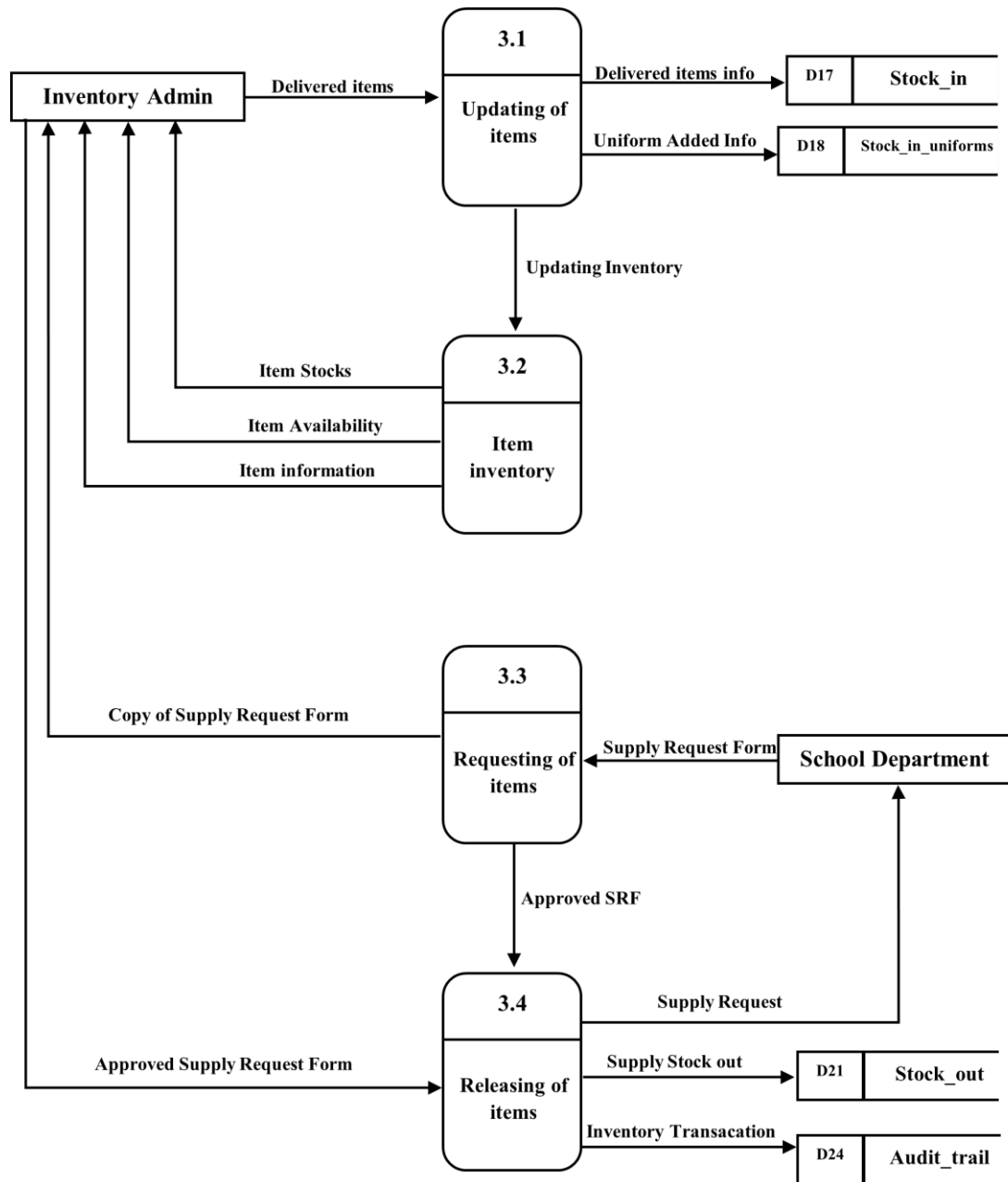


Figure 12: Diagram 1 of Process 3 (Proposed)

Figure 12 the data flow diagram of process 3 of the proposed system of General Services Department. Once the inventory admin receives the delivered items and performs a thorough check to ensure the accuracy and quality of the items. The inventory admin can now update the inventory records to reflect the new stock. This includes recording the quantities, item details, and other relevant information in the inventory. With the inventory restocked, the General Services Department can now facilitate the process of fulfilling requests from students and school departments for supplies. All database that is connected to the inventory module will update immediately after the transactions, this includes stock in, stock out, school department, and audit trail.

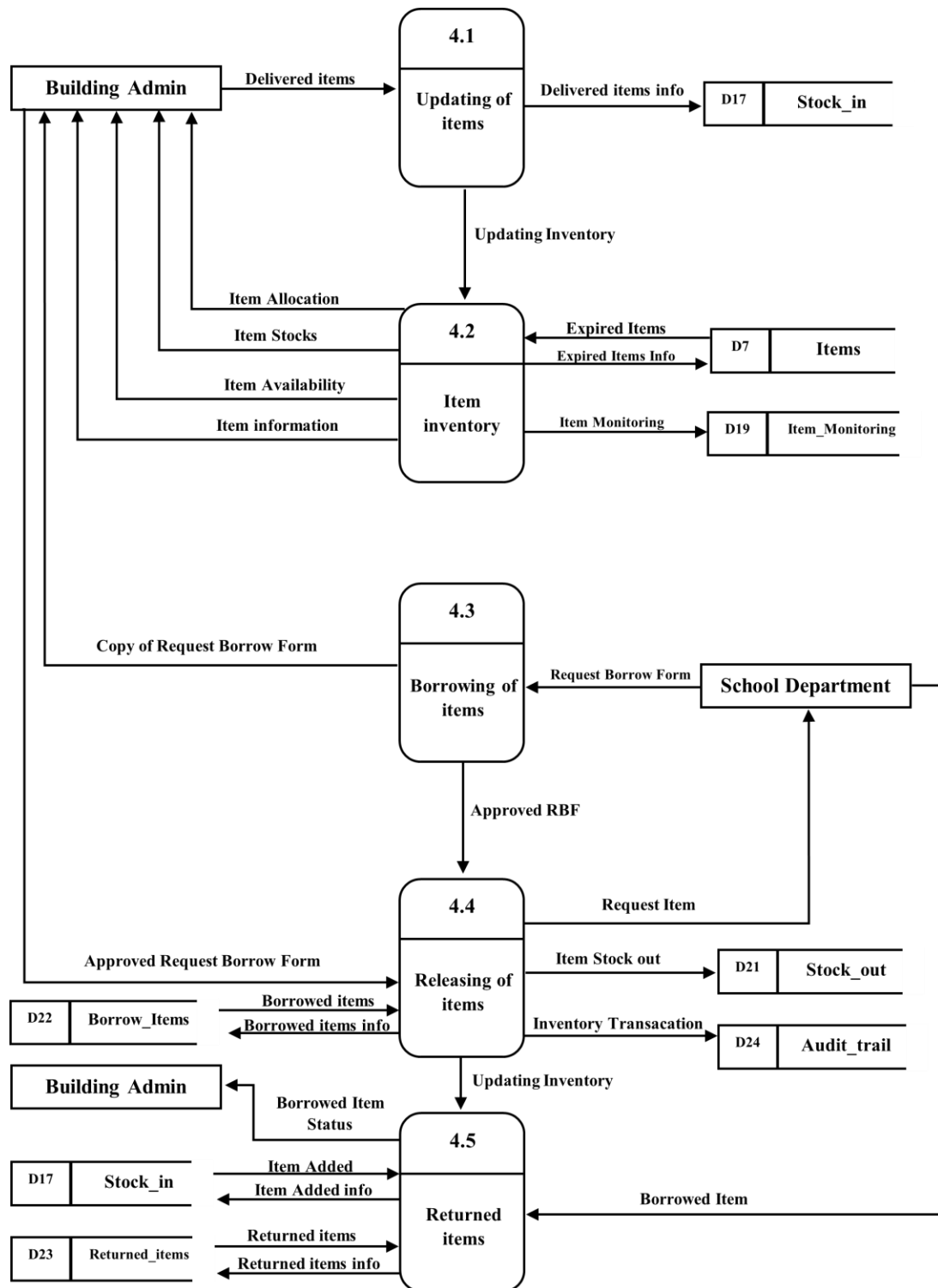


Figure 13: Diagram 1 of Process 4 (Proposed)

Figure 13 the data flow diagram of process 4 of the proposed system of General Services Department. Once the building admin receives the delivered items and performs any necessary checks or inspections to ensure the quality and correctness of the items. The building admin can now update the inventory records to reflect the new stock. This includes recording the quantities, item details, and other relevant information in the inventory. After the inventory for building resources is restocked, the General Services Department can now facilitate the process of fulfilling requests from students and school departments to borrow items for use inside the school. Students and school departments can now submit their requests to the General Services Department, specifying the items they need to borrow for their respective purposes within the school. As soon as the transactions on the borrowing of items are done the system will update the stocks and information related to the building module immediately. After the students or school departments have finished using the borrowed items, they return them to the General Services Department. After the building admin receives the returned items and performs necessary inspections the system can now update and record the informations within the building module.

3.6 ACTIVITY DIAGRAM

An activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. Activity diagrams may be regarded as a form of flowchart.

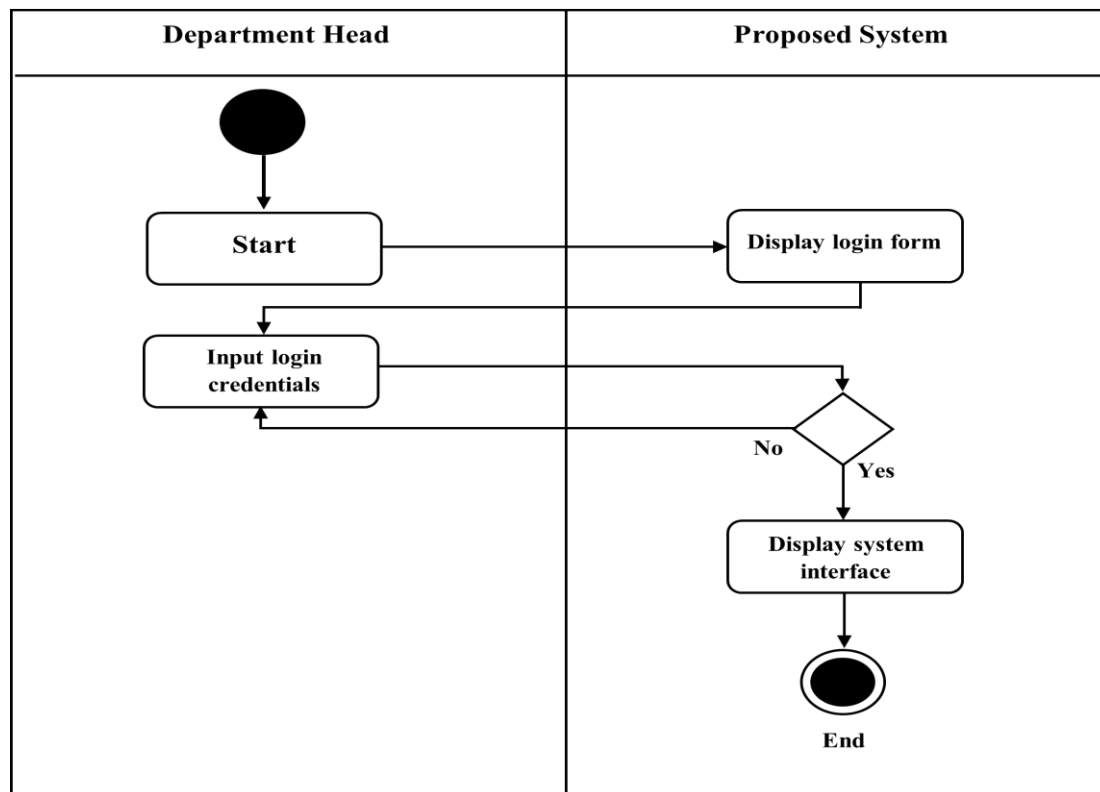


Figure 14: Activity Diagram (Login)

Figure 14 Activity Diagram (Login) by the Department Head shows the login process of the system. If the entered credentials are incorrect, the system will display an error message indicating that the login credentials are incorrect. After displaying the error message, the system typically provides options for the user, such as retrying the login process or exiting the system. If the user chooses to retry, the system goes back to the user input step. If the user chooses to exit, the activity diagram ends. And if the login credentials of the user are correct, it will lead the user to the system.

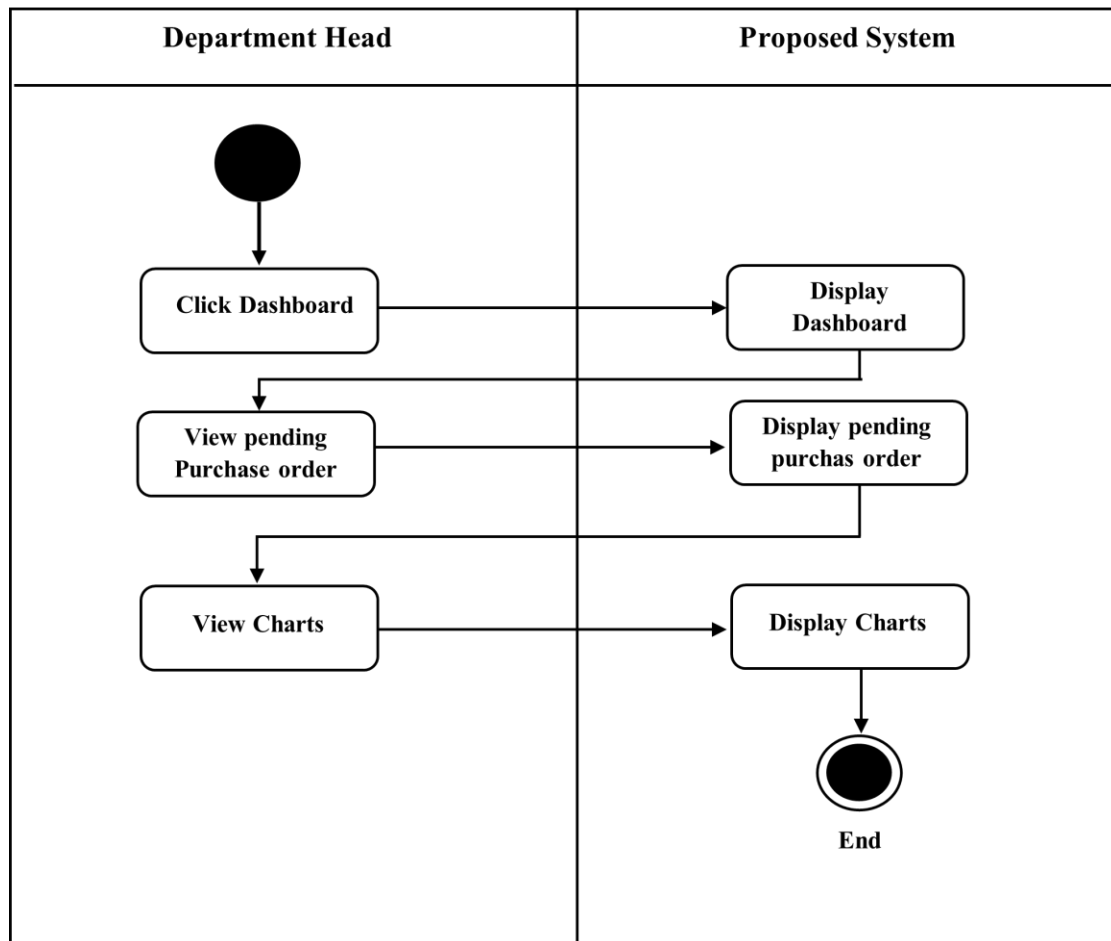


Figure 15: Activity Diagram (Dashboard)

Figure 15 Activity Diagram (Dashboard) by the Department Head shows the next process of the system, where it displays the dashboard of the system. The dashboard includes charts and the status of the purchase orders.

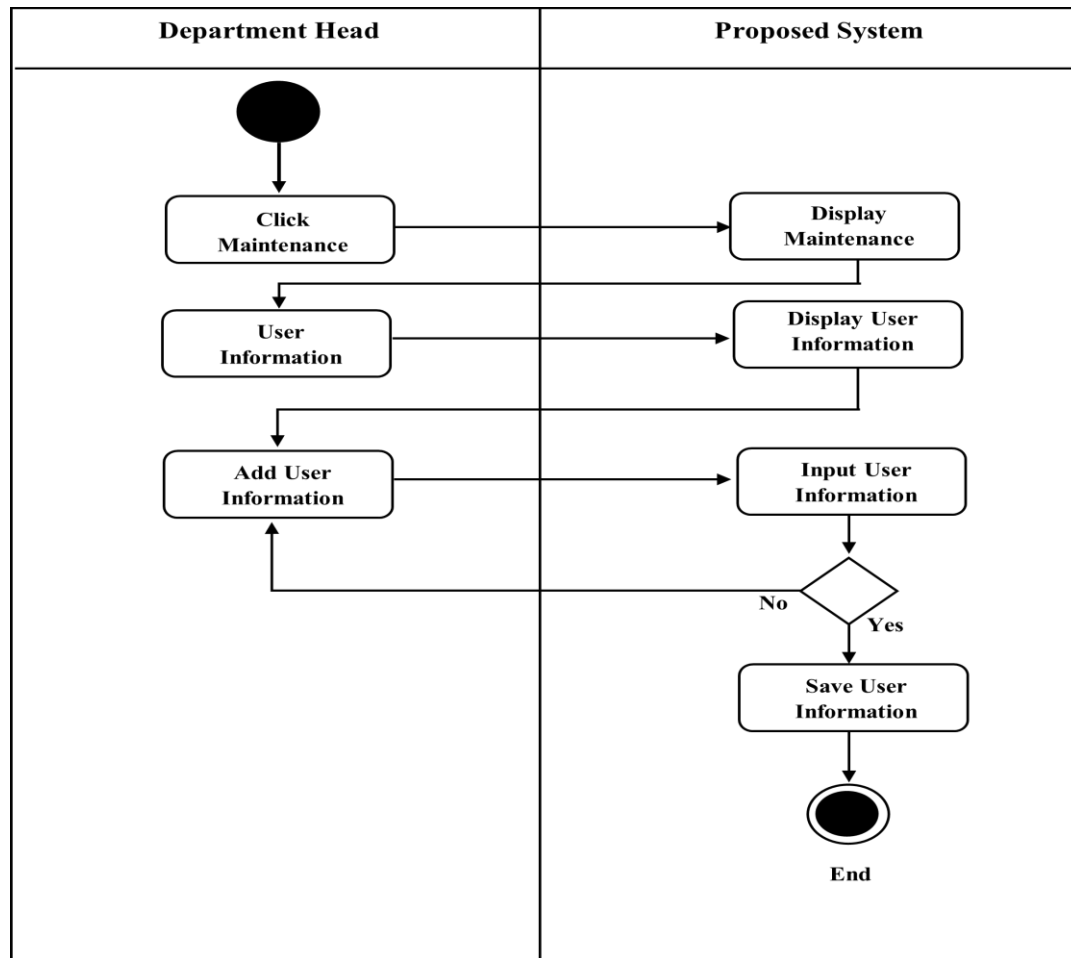


Figure 16: Activity Diagram (Maintenance & User Information)

Figure 16 Activity Diagram (Maintenance & User Information). The system prompts the department head to provide the necessary information for adding a new user. If the entered information is valid, the system proceeds to add the user to the system after the department head added the user information the system will updates the user database with the new user information.

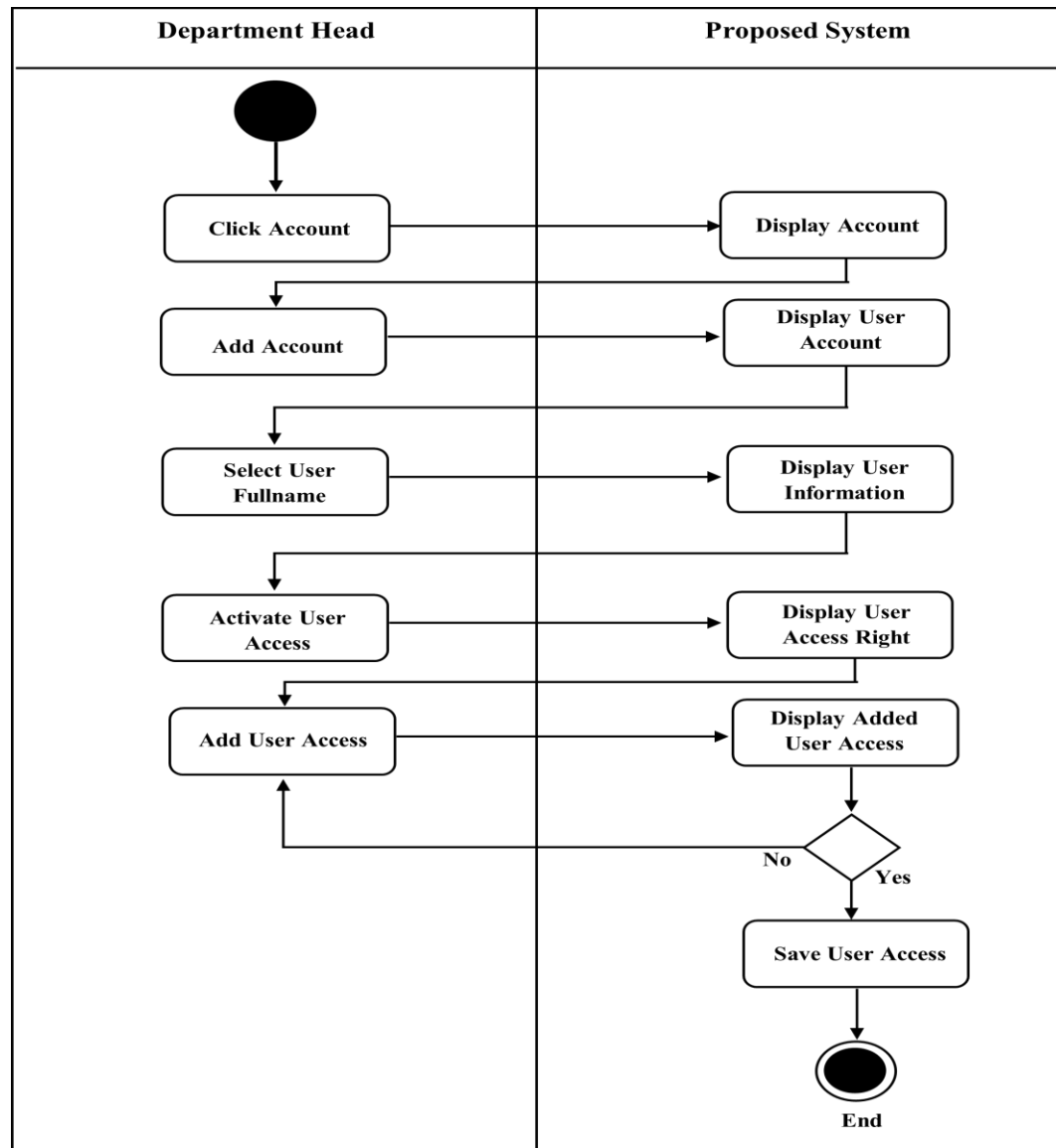


Figure 17: Activity Diagram (Account)

Figure 17 Activity Diagram (Account) shows the process of giving the user access to the system. If the identification information is valid, the system proceeds to grant access to the user. After that the system updates the user's access database in the system based on their role or assigned access level.

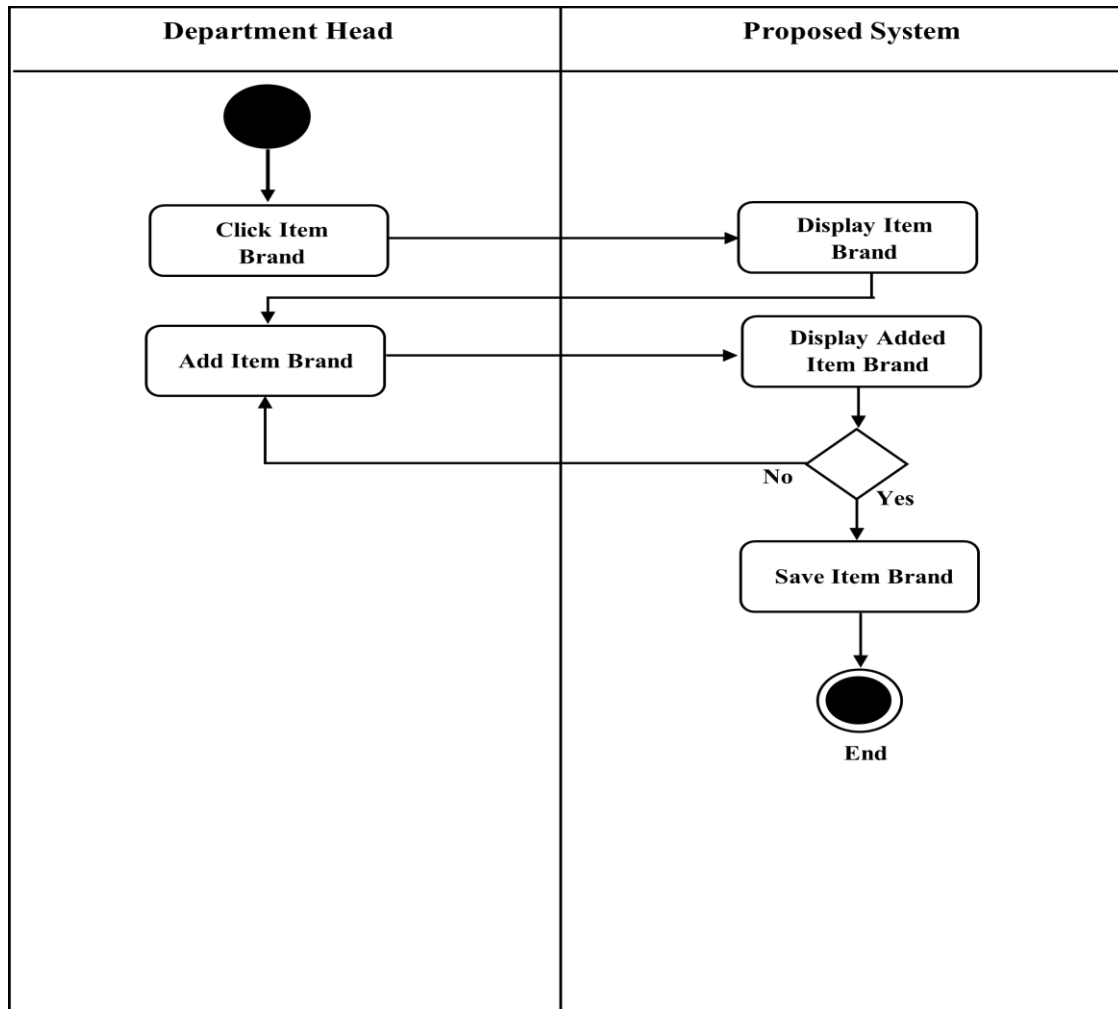


Figure 18: Activity Diagram (Item Brand)

Figure 18 Activity Diagram (Item Brand) shows the process of adding item brands on the system where it gets stored on the systems database.

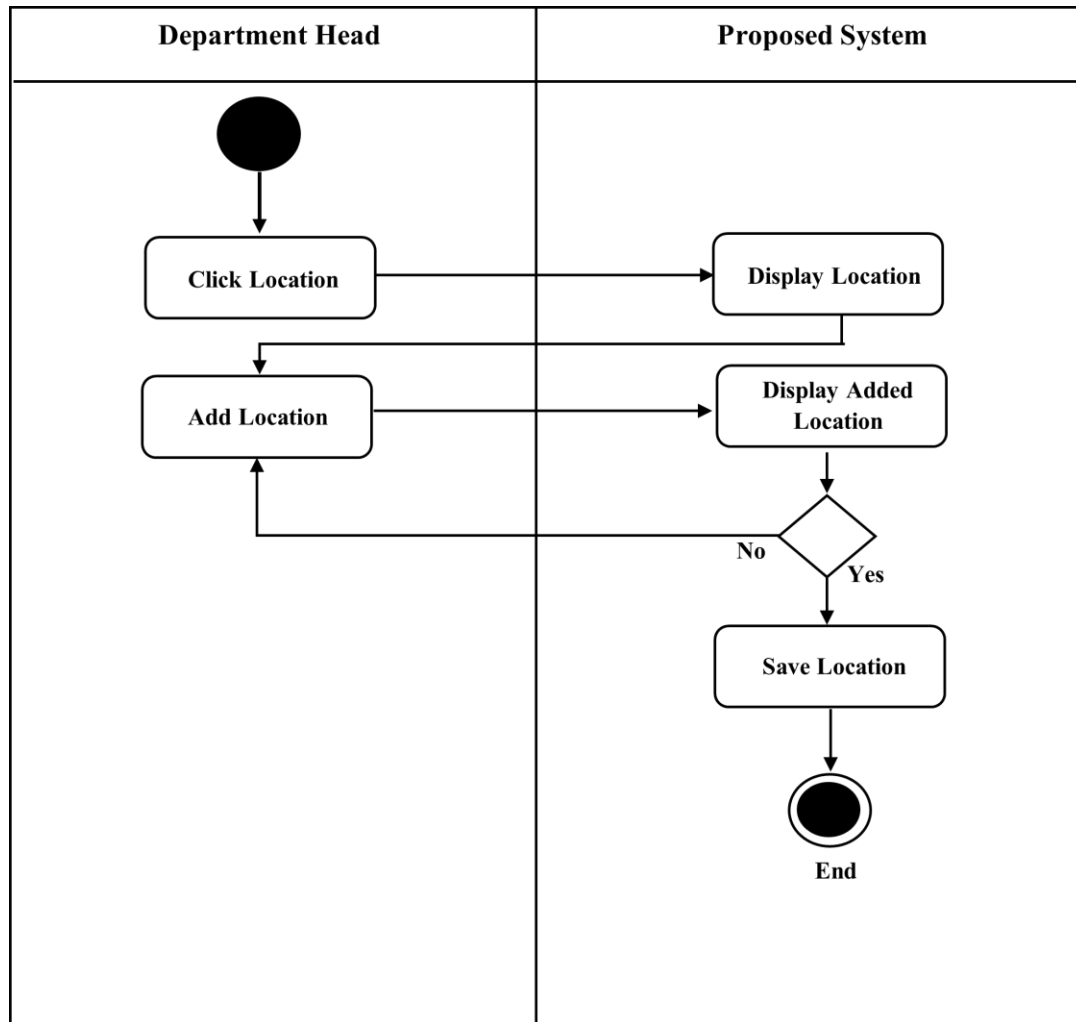


Figure 19: Activity Diagram (Location)

Figure 19 Activity Diagram (Location) shows the process of adding location on the system where it gets stored on the systems database.

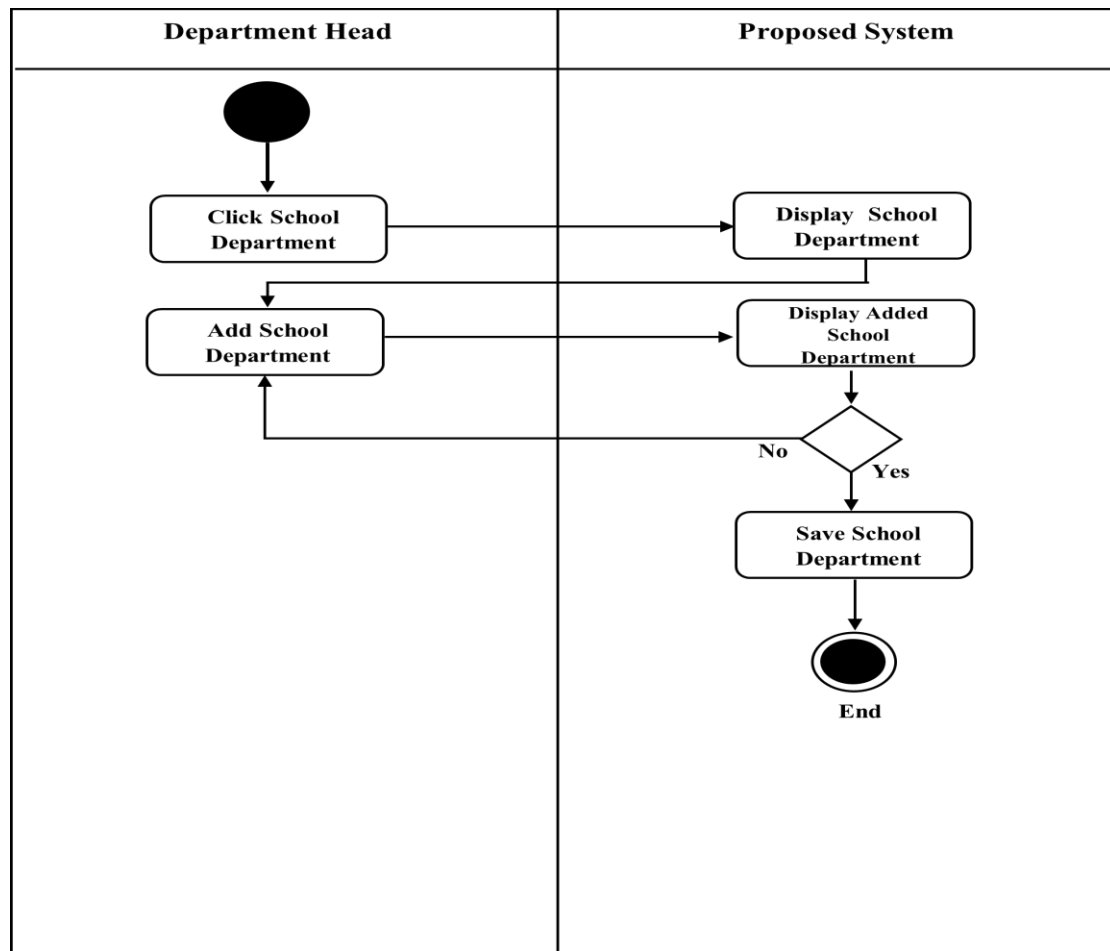


Figure 20: Activity Diagram (School Department)

Figure 20 Activity Diagram (School Department) shows the process of adding a school department to the system and how it gets stored in the system database. The school department is the one that requests and borrows the items from the GS department.

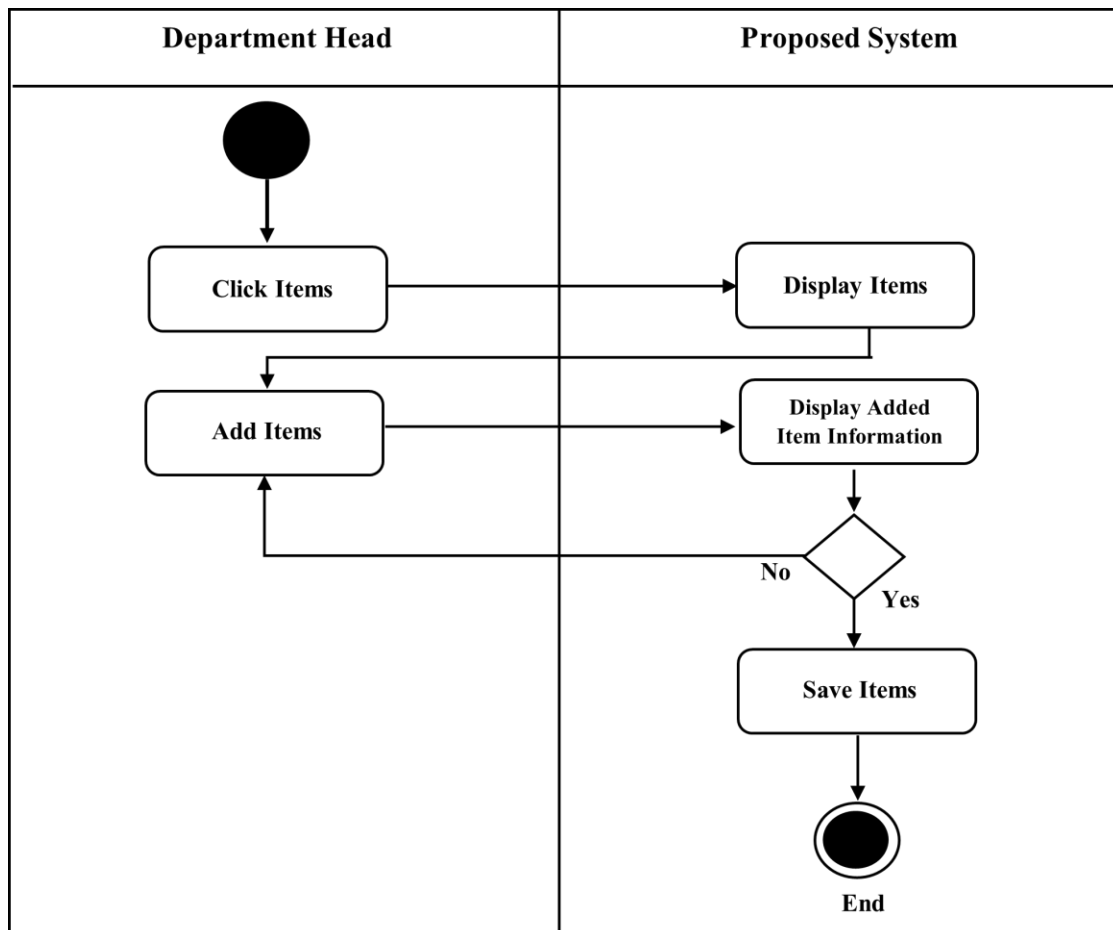


Figure 21: Activity Diagram (Items)

Figure 21 Activity Diagram (Items) shows the process of adding item informations to the system and how it gets stored in the system database.

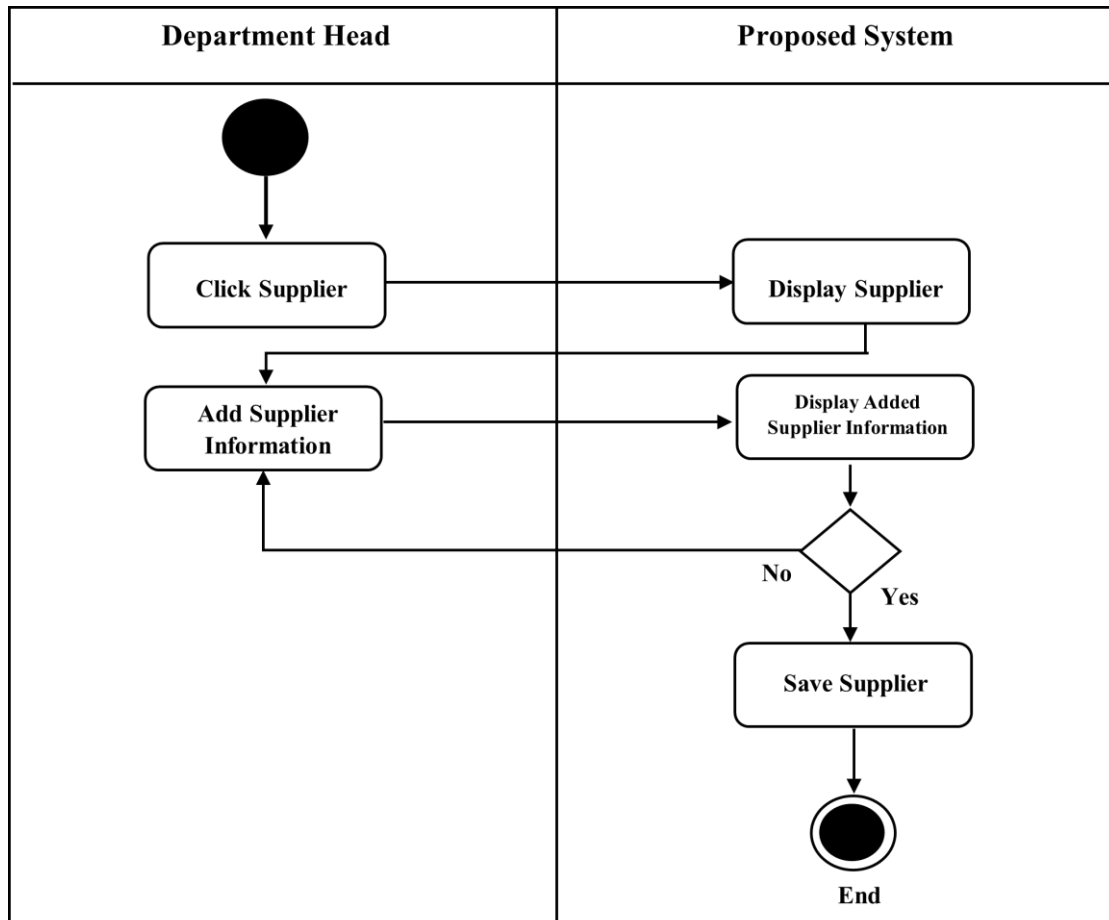


Figure 22: Activity Diagram (Supplier)

Figure 22 Activity Diagram (Supplier) shows the process of adding the supplier's informations to the system and how it gets stored in the system database.

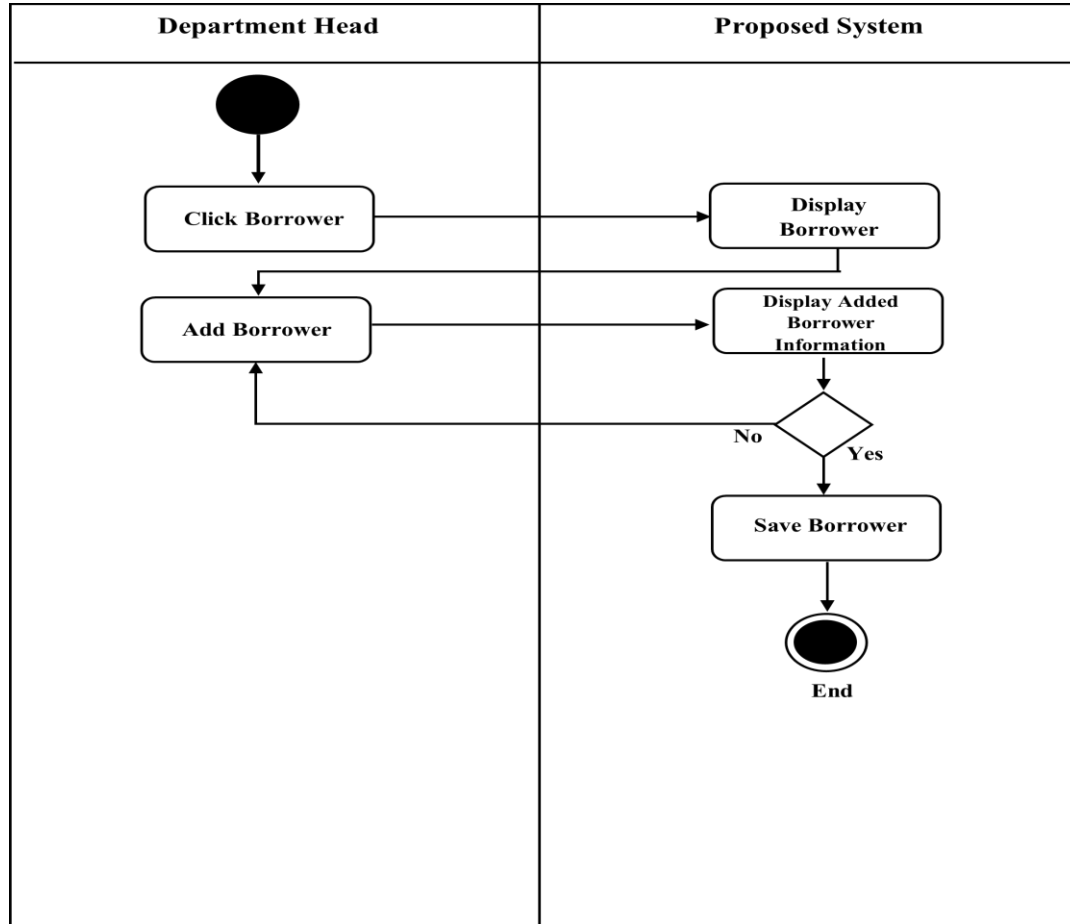


Figure 23: Activity Diagram (Borrower)

Figure 23 Activity Diagram (Borrower) shows the process of adding borrower informations to the system and how it gets stored in the system database.

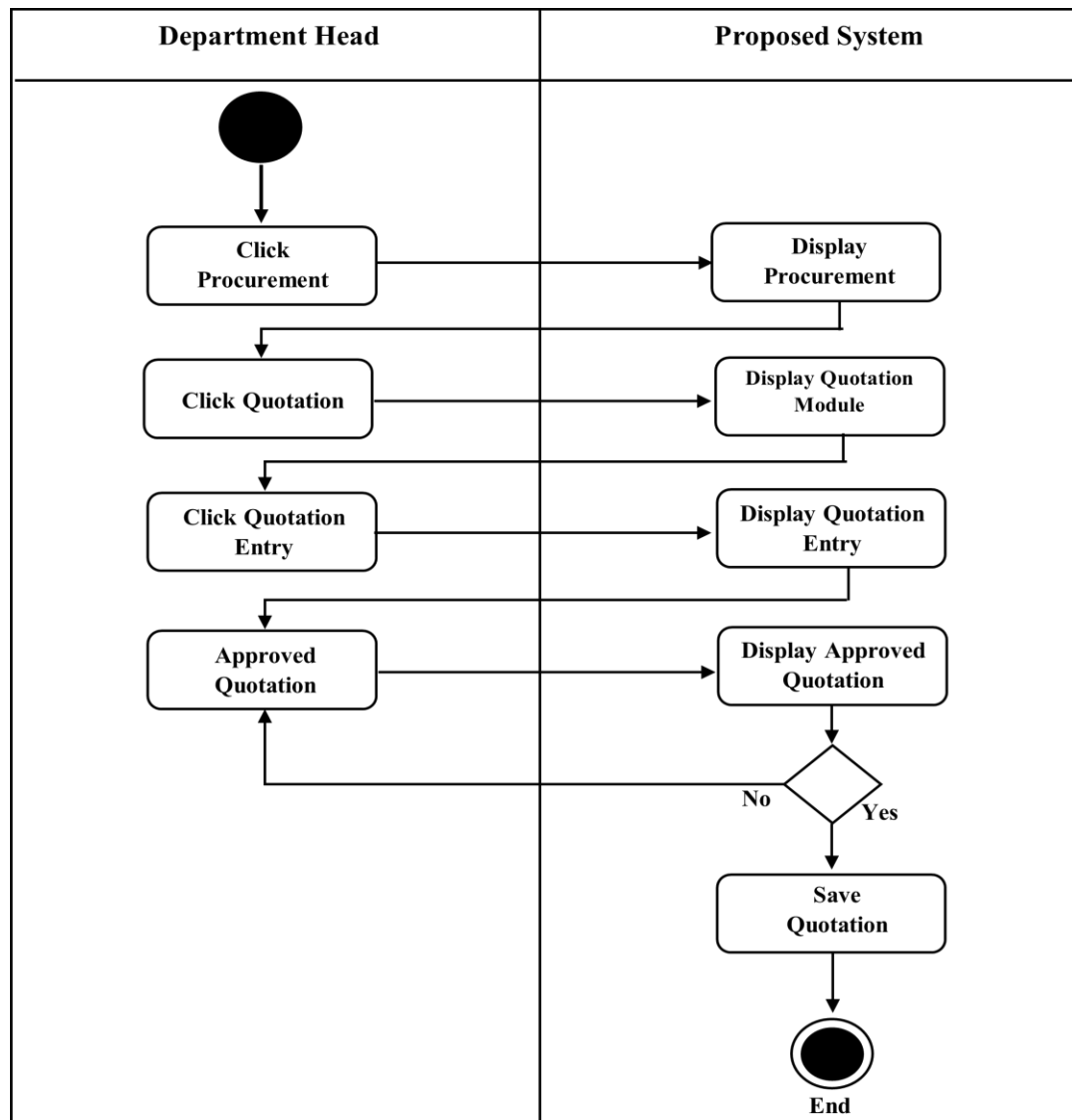


Figure 24: Activity Diagram (Procurement & Quotations)

Figure 24 Activity Diagram (Procurement & Quotations) shows the process of checking and approving of quotation entries on the system and how it gets stored in the system database. The department head is the one who is responsible for checking the quotations submitted by the department staffs.

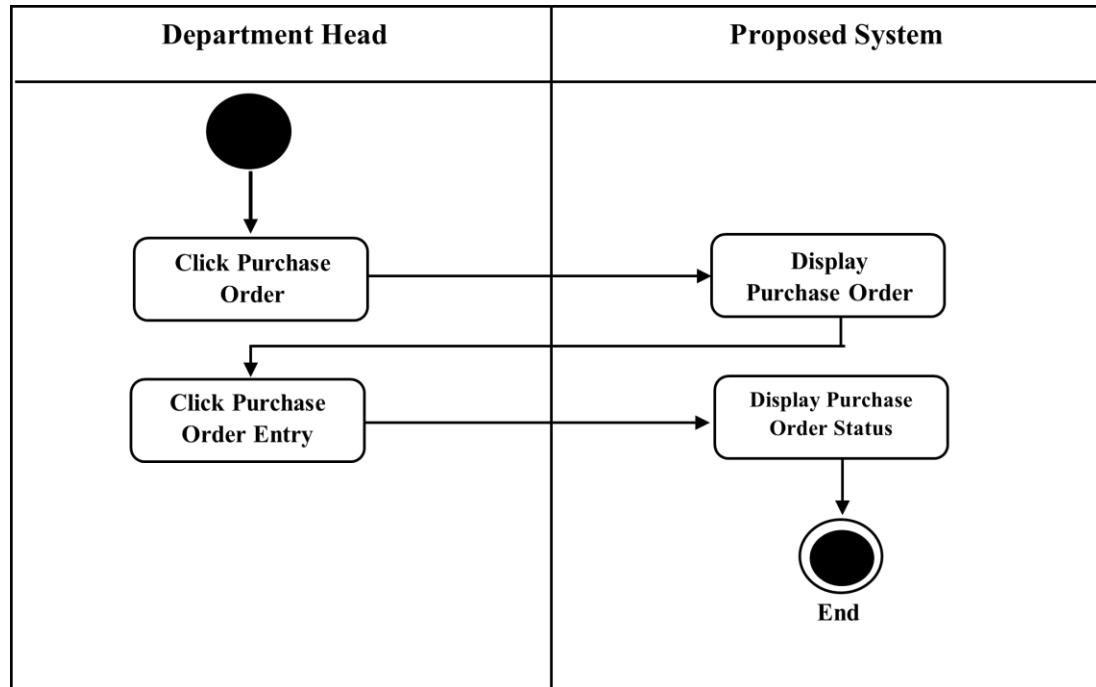


Figure 25: Activity Diagram (Purchase Order)

Figure 25 Activity Diagram (Purchase Order) shows you the process where the department head checks the status of the purchase orders of the department.

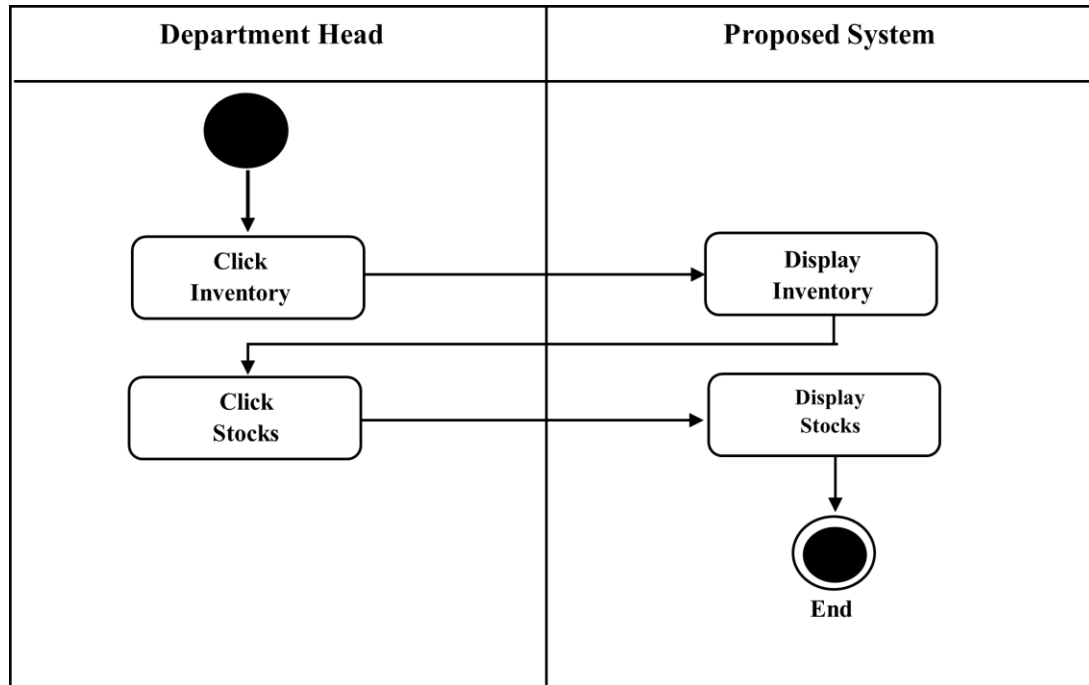


Figure 26: Activity Diagram (Inventory Monitoring)

Figure 26 Activity Diagram (Inventory Monitoring) is the process where the department head monitors the inventory of the department. The head checks the item stocks and item availability.

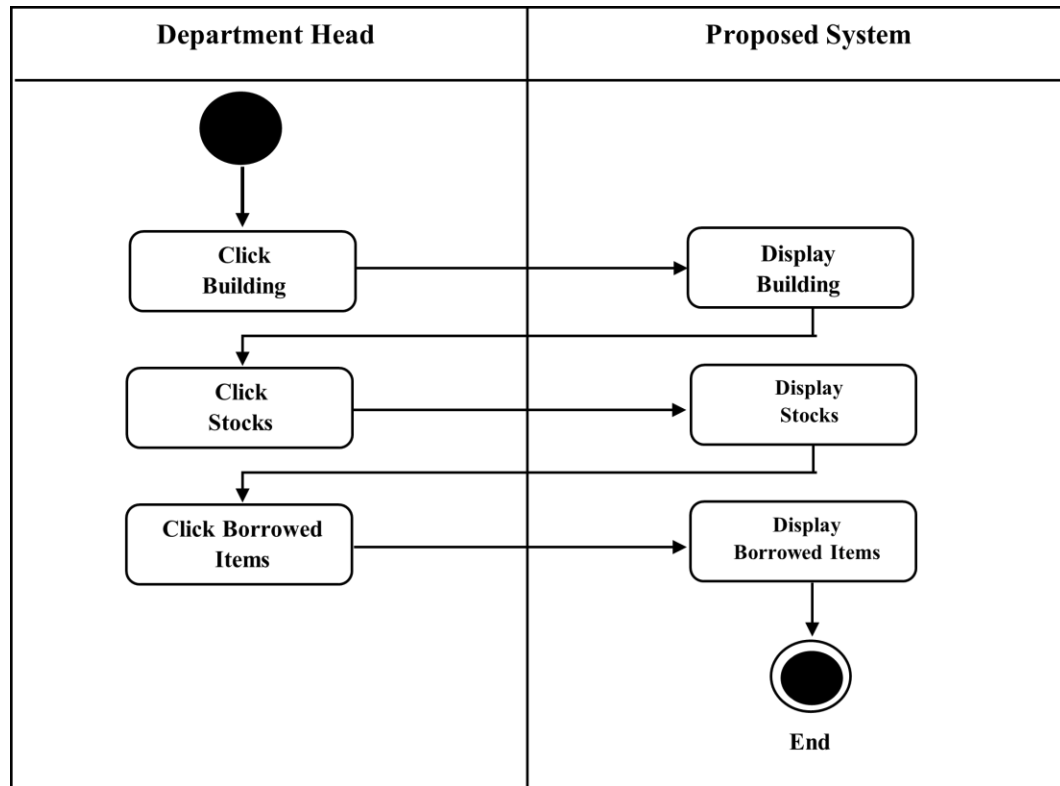


Figure 27: Activity Diagram (Building Monitoring)

Figure 27 Activity Diagram (Building Monitoring) is the process where the department head monitors the inventory of the building admin. The head checks the item stocks, availability and if the items are borrowed, damage, and expired.

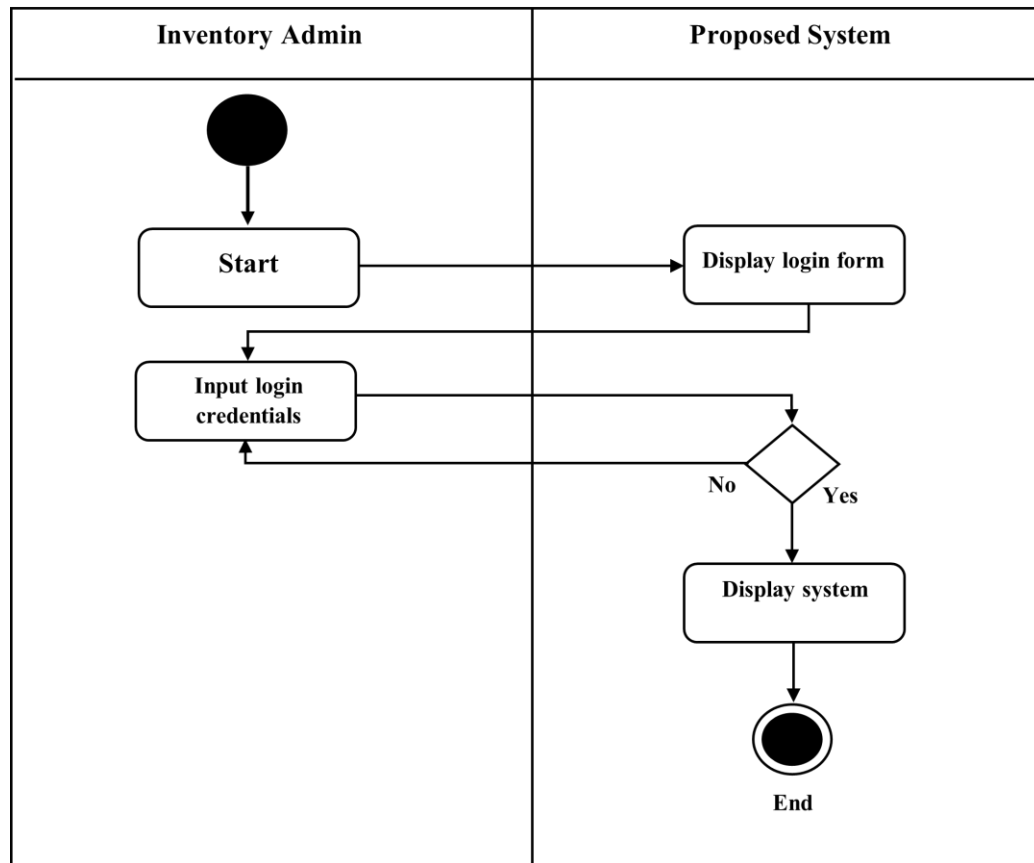


Figure 28: Activity Diagram (Inventory Admin Login)

Figure 28 Activity Diagram (Inventory Admin Login). This diagram shows the login process of the inventory admin to the system.

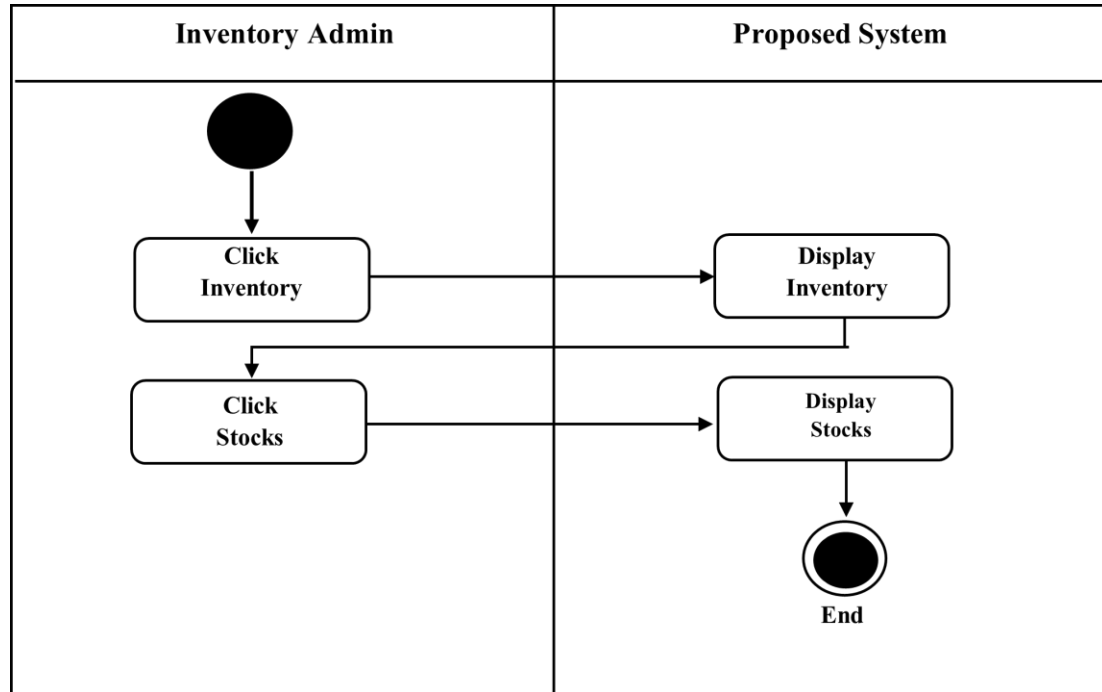


Figure 29: Activity Diagram (Inventory)

Figure 29 Activity Diagram (Inventory) is the process where the inventory admin monitors the inventory of the department to check if the items are still available or if the items are out-of-stock.

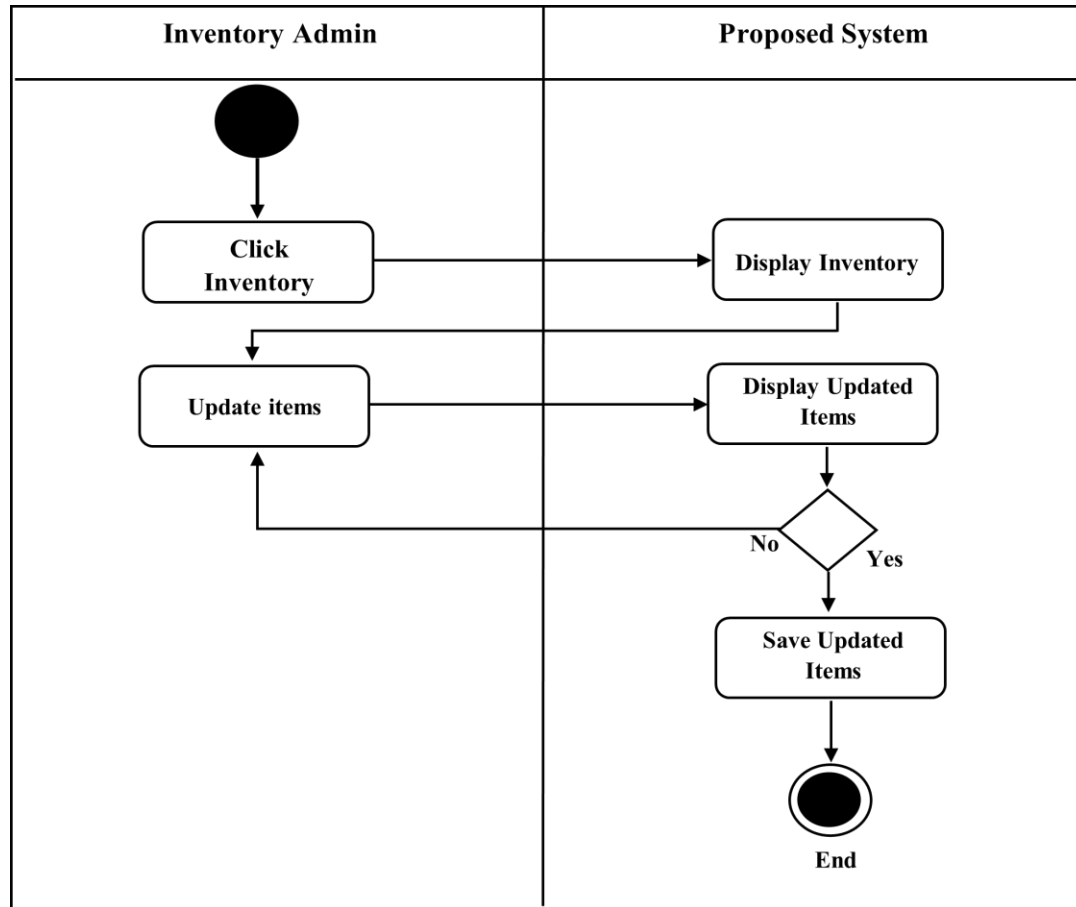


Figure 30: Activity Diagram (Updating of Inventory)

Figure 30 Activity Diagram (Updating of Inventory) is the process where the inventory admin updates the inventory of the department after the items are delivered to them.

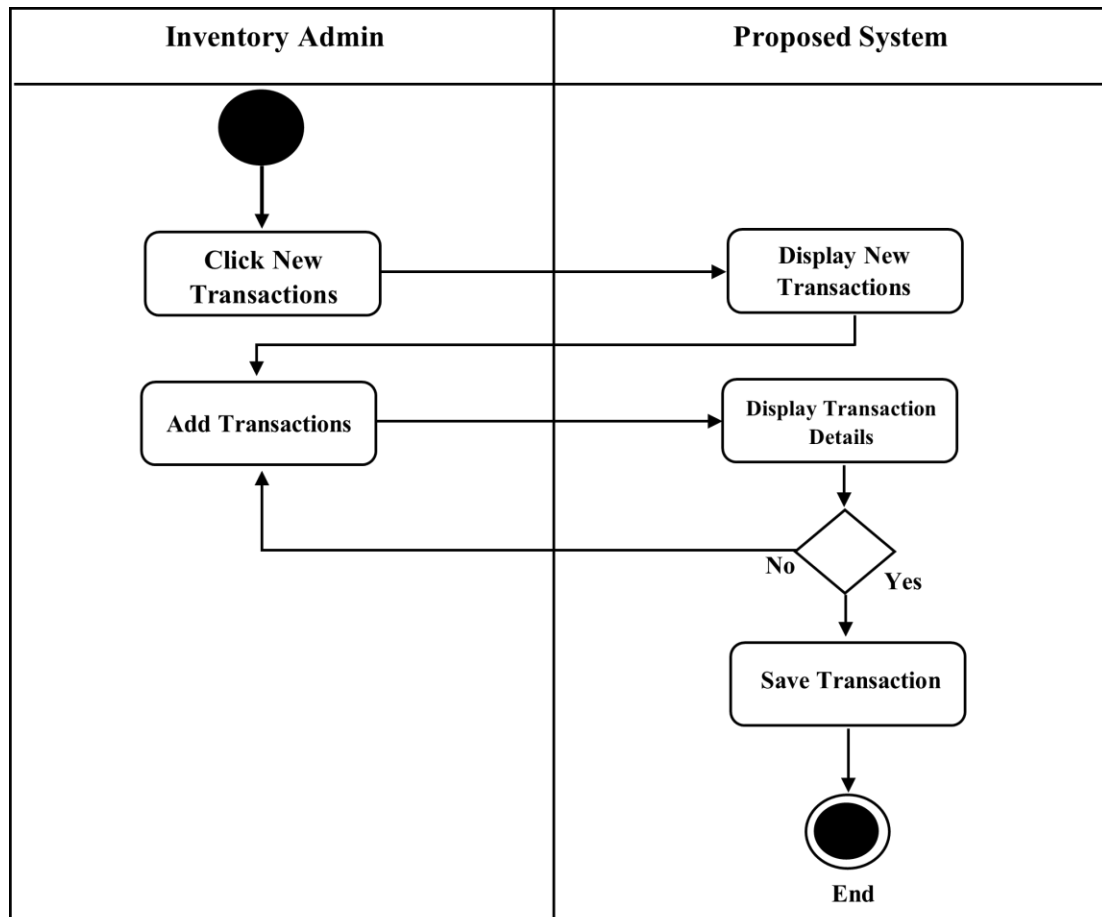


Figure 31: Activity Diagram (New Transaction (SRF))

Figure 31 Activity Diagram (New Transaction (SRF)) is the process where the inventory admin adds a transaction when school departments are requesting supplies from them.

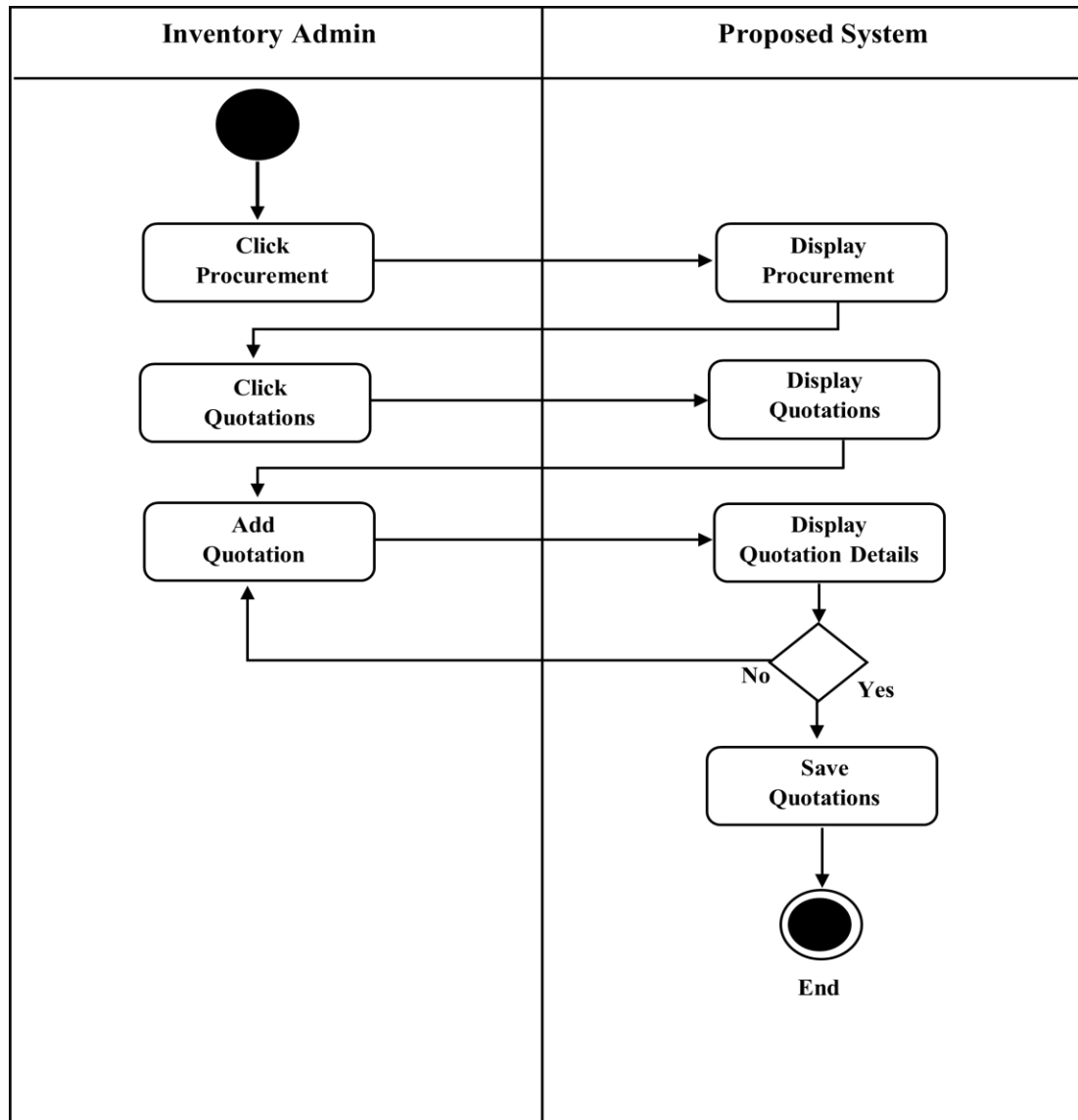


Figure 32: Activity Diagram (Inventory Quotation)

Figure 32 Activity Diagram (Inventory Quotation) is the process where the inventory admin handed his/her quotation of the items needed to be restock to the system.

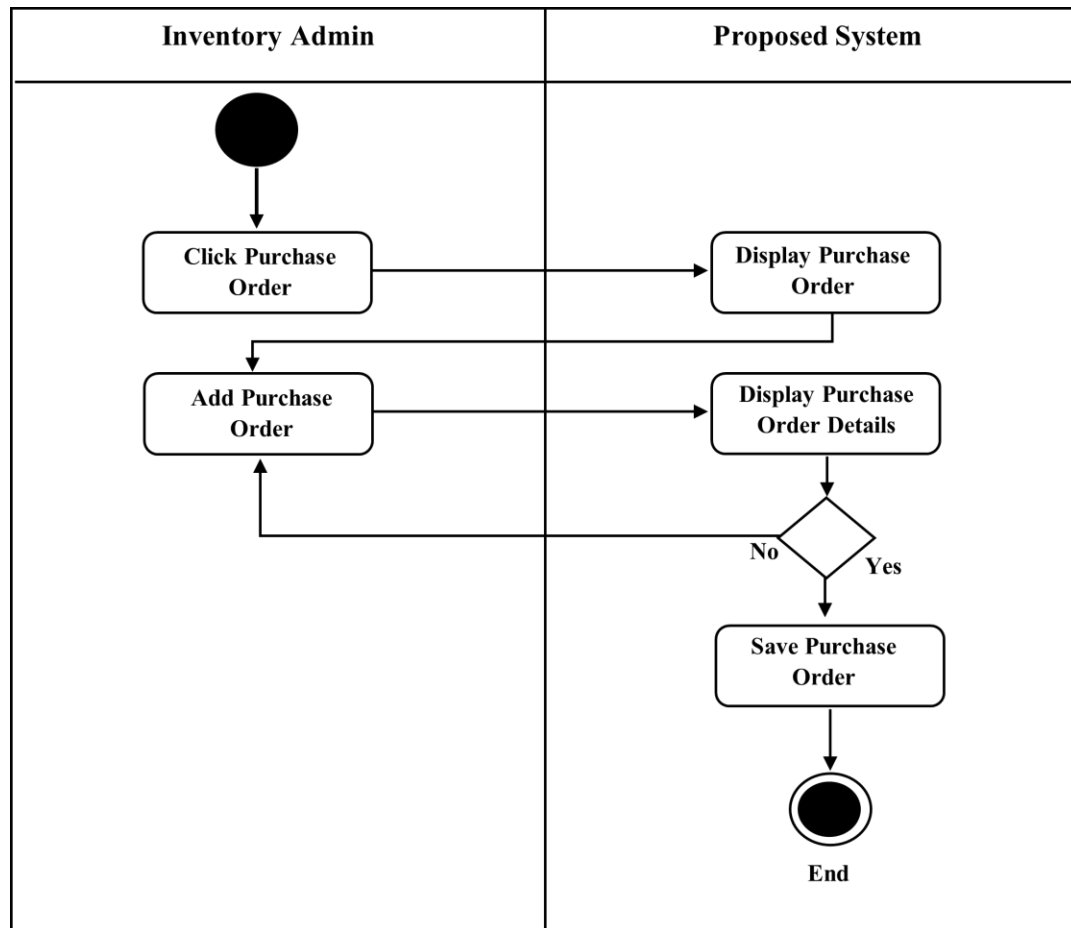


Figure 33: Activity Diagram (Inventory Purchase Order)

Figure 33 Activity Diagram (Inventory Purchase Order) shows you the process of adding a purchase order entry of the inventory admin after the quotation was approved by the department head.

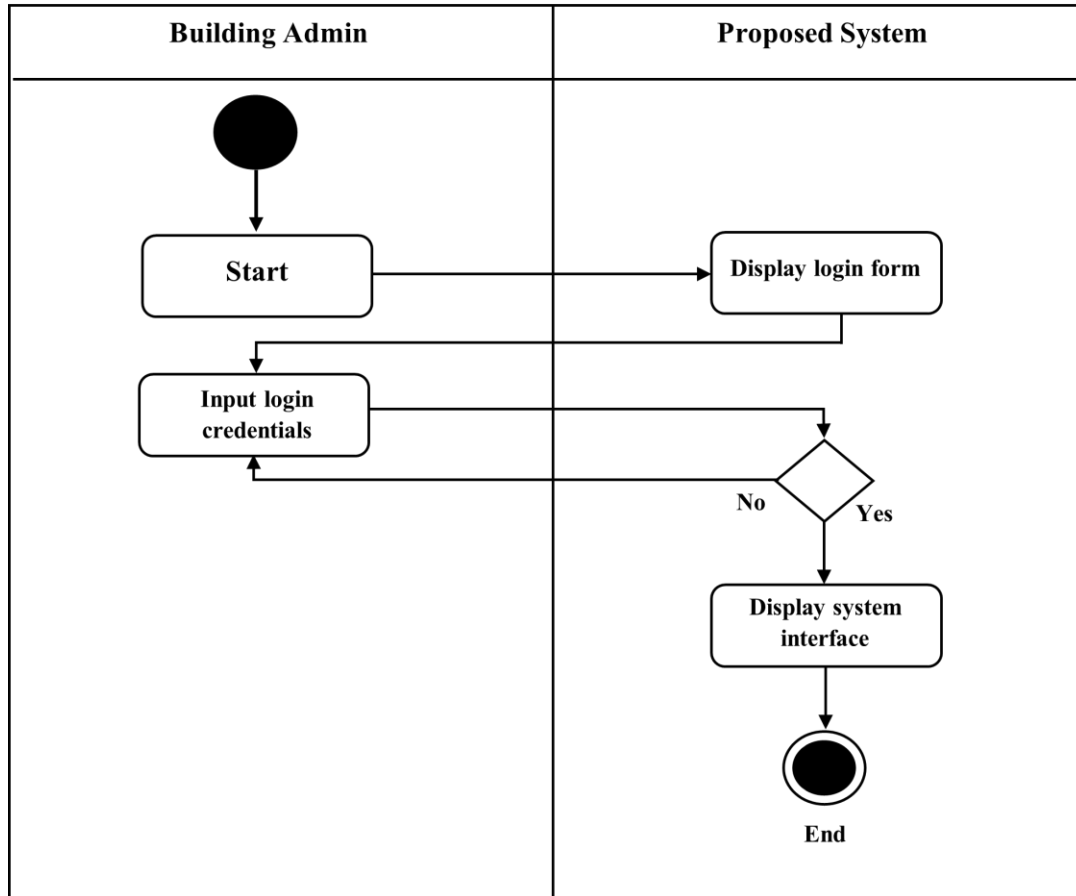


Figure 34: Activity Diagram (Building Admin Login)

Figure 34 Activity Diagram (Building Admin Login). This diagram shows the login process of the building admin to the system.

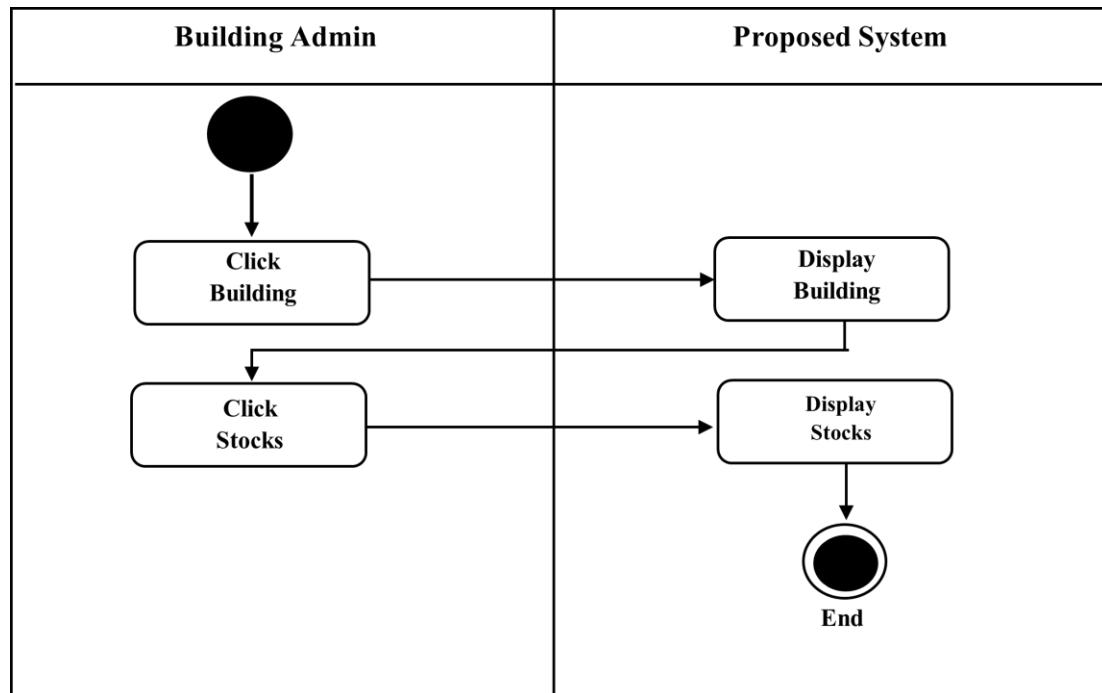


Figure 35: Activity Diagram (Building)

Figure 35 Activity Diagram (Building) is the process where the building admin monitors the building inventory of the department to check if the items are still available, borrowed or if the items are out-of-stock.

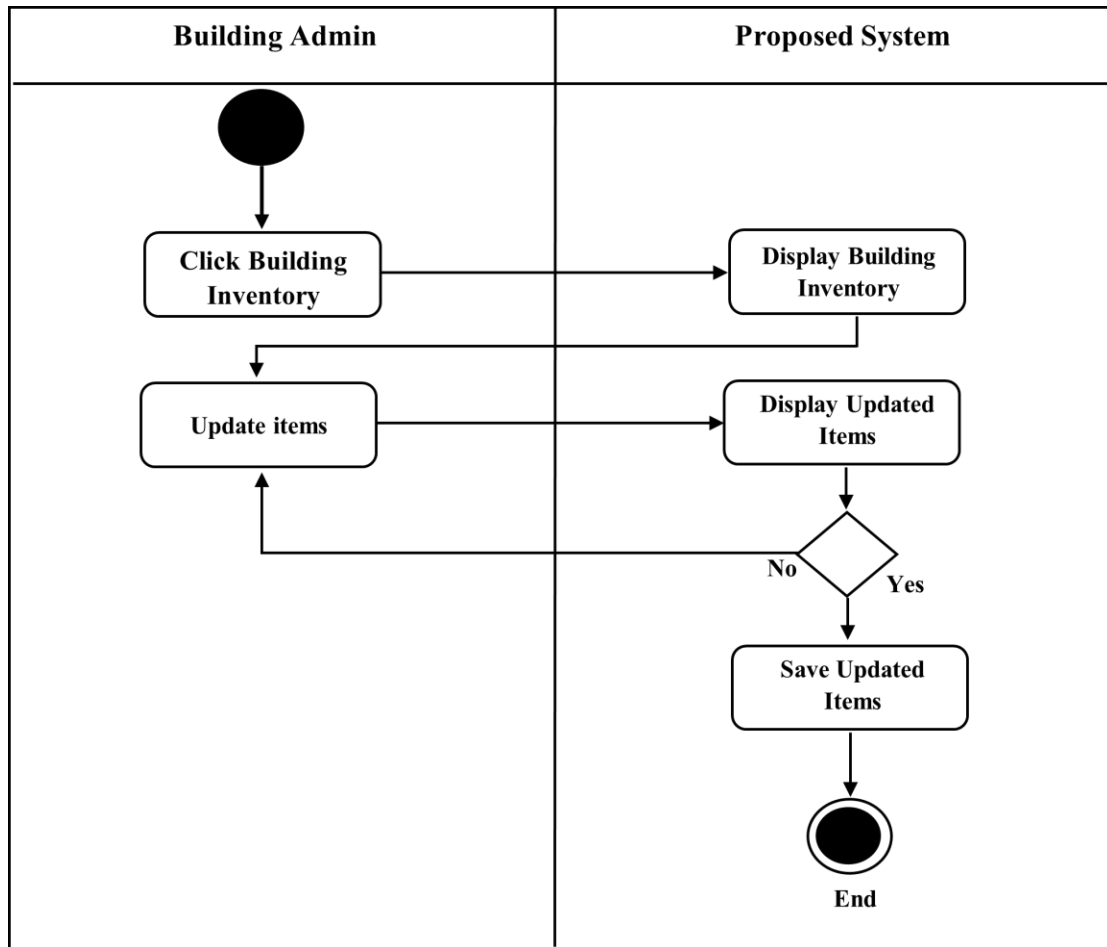


Figure 36: Activity Diagram (Updating of Building Inventory)

Figure 36 Activity Diagram (Updating of Building Inventory) is the process where the building admin updates the building inventory of the department after the items are delivered to them.

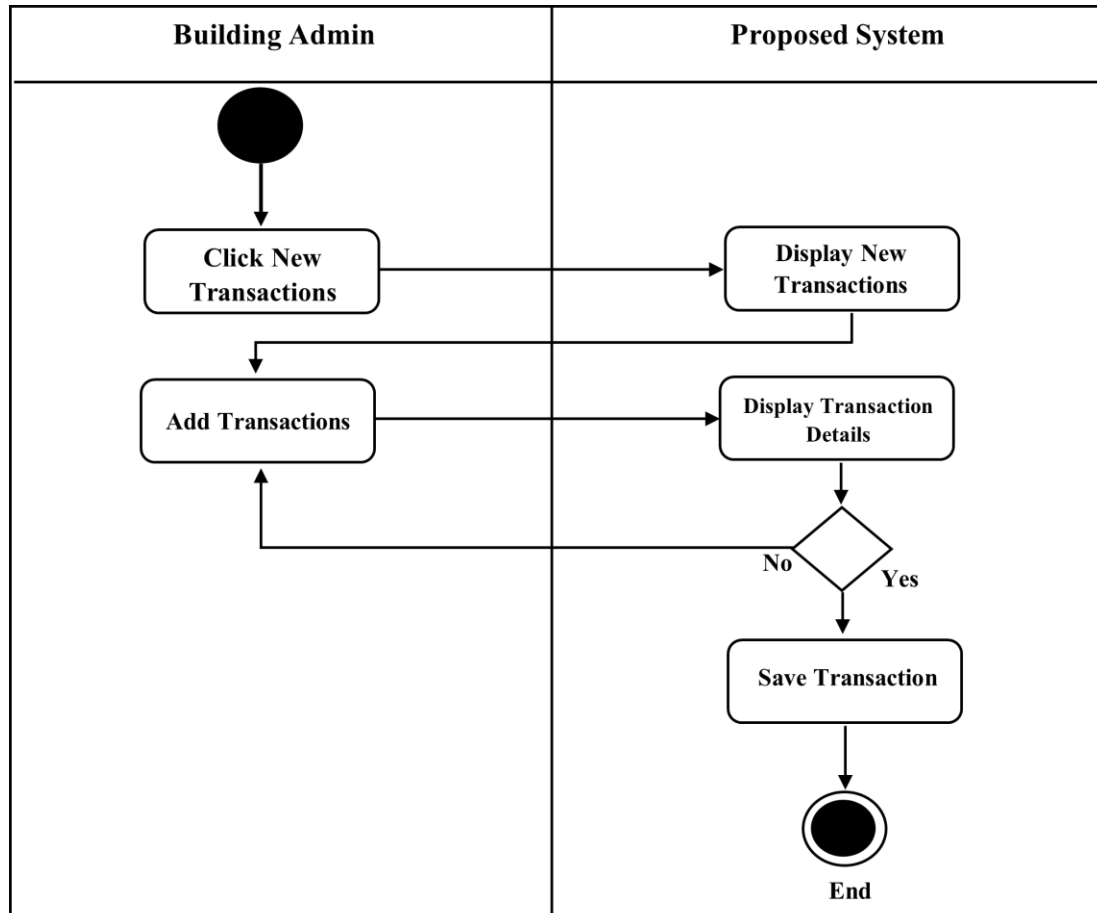


Figure 37: Activity Diagram (New Transaction (RBF))

Figure 37 Activity Diagram (New Transaction (RBF)) is the process where the building admin adds a transaction when school departments are borrowing supplies from them.

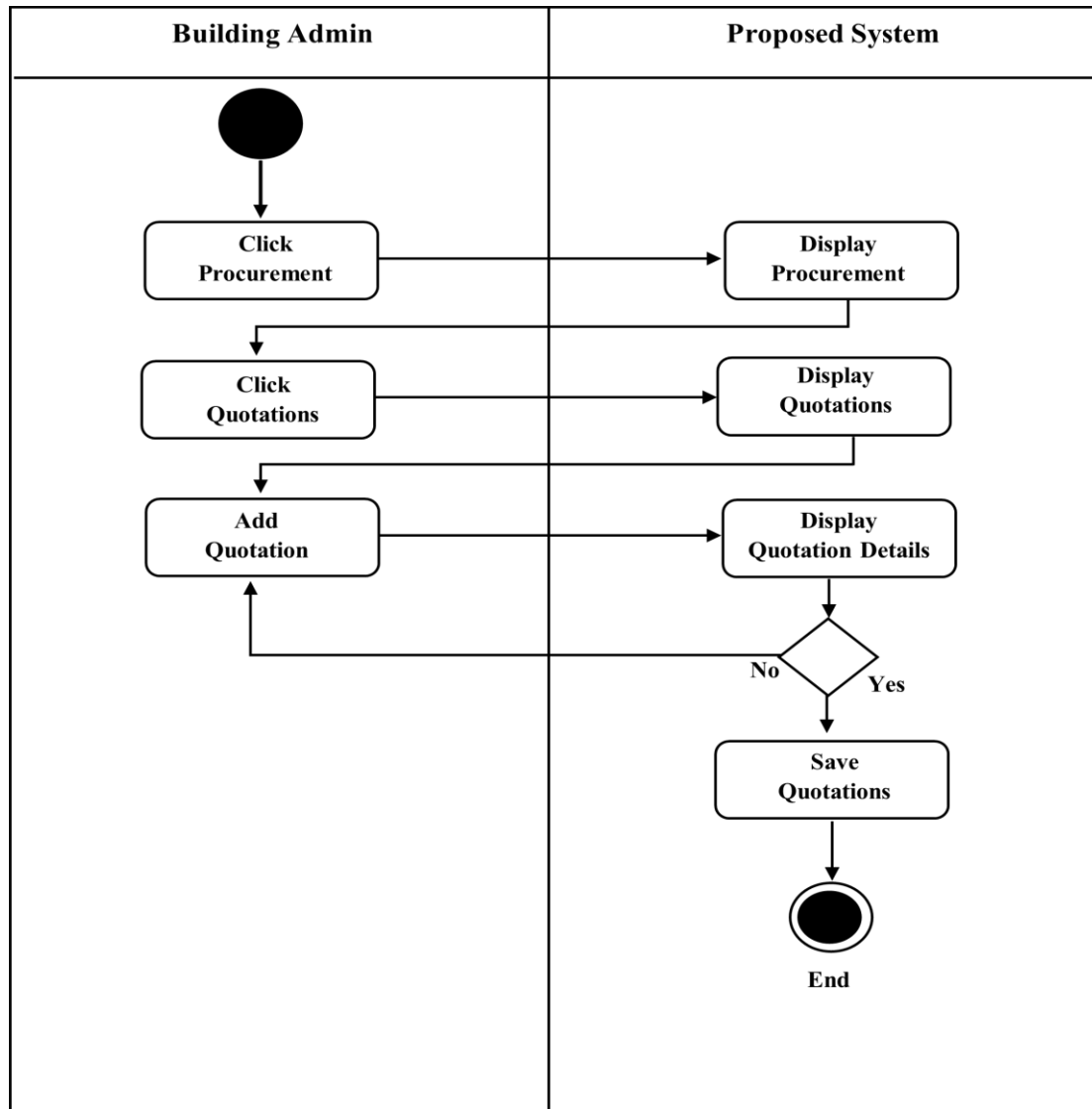


Figure 38: Activity Diagram (Building Quotation)

Figure 38 Activity Diagram (Building Quotation) is the process where the inventory admin handed his/her quotation of the items needed to be restock to the system.

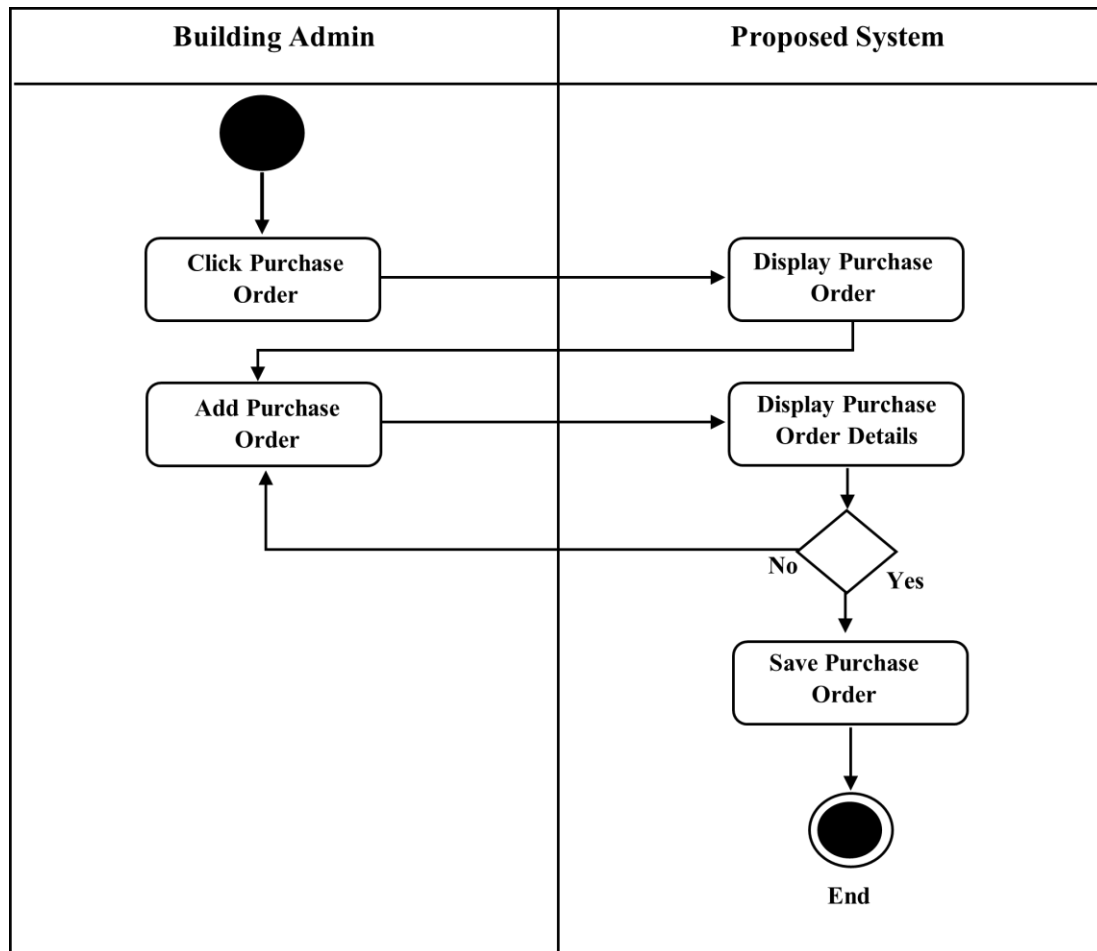


Figure 39: Activity Diagram (Building Purchase Order)

Figure 39 Activity Diagram (Building Purchase Order) shows you the process of adding a purchase order entry of the building admin after the quotation was approved by the department head.

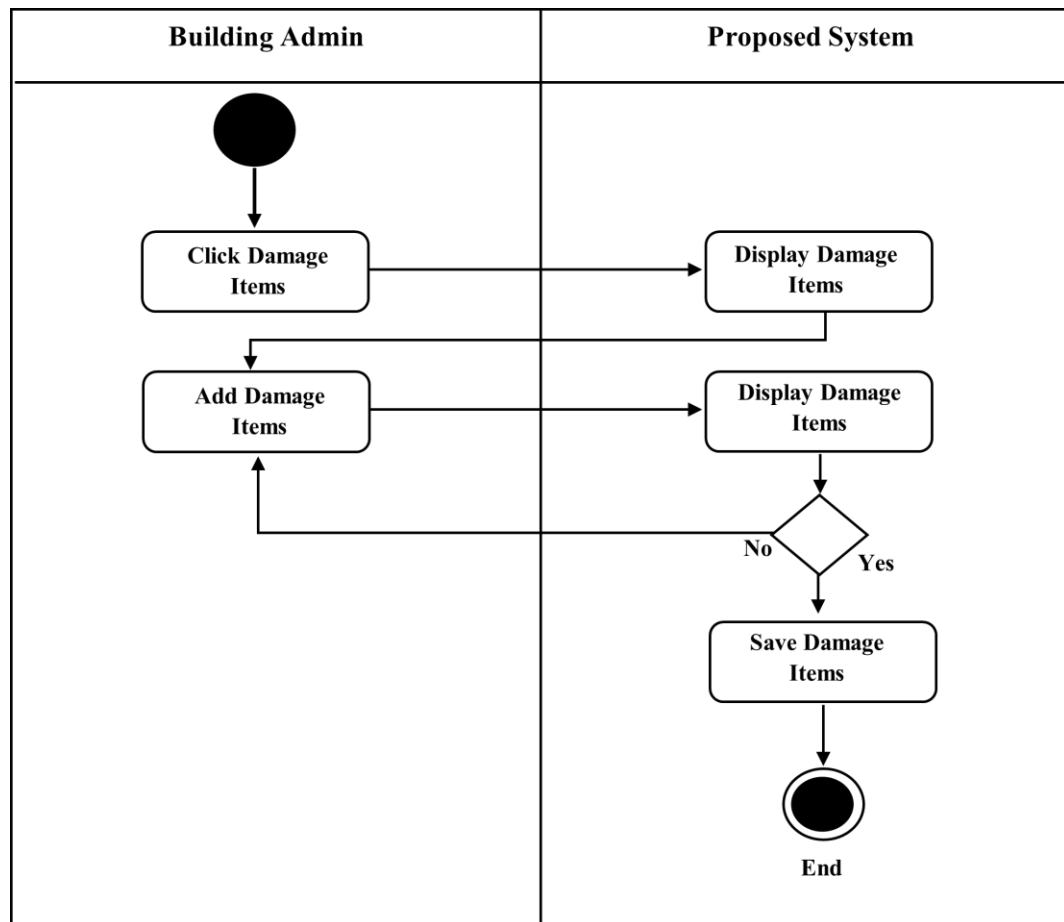


Figure 40: Activity Diagram (Damage Items)

Figure 40 Activity Diagram (Damage Items) shows the process of adding damage report to the system and how it gets stored to the systems database.

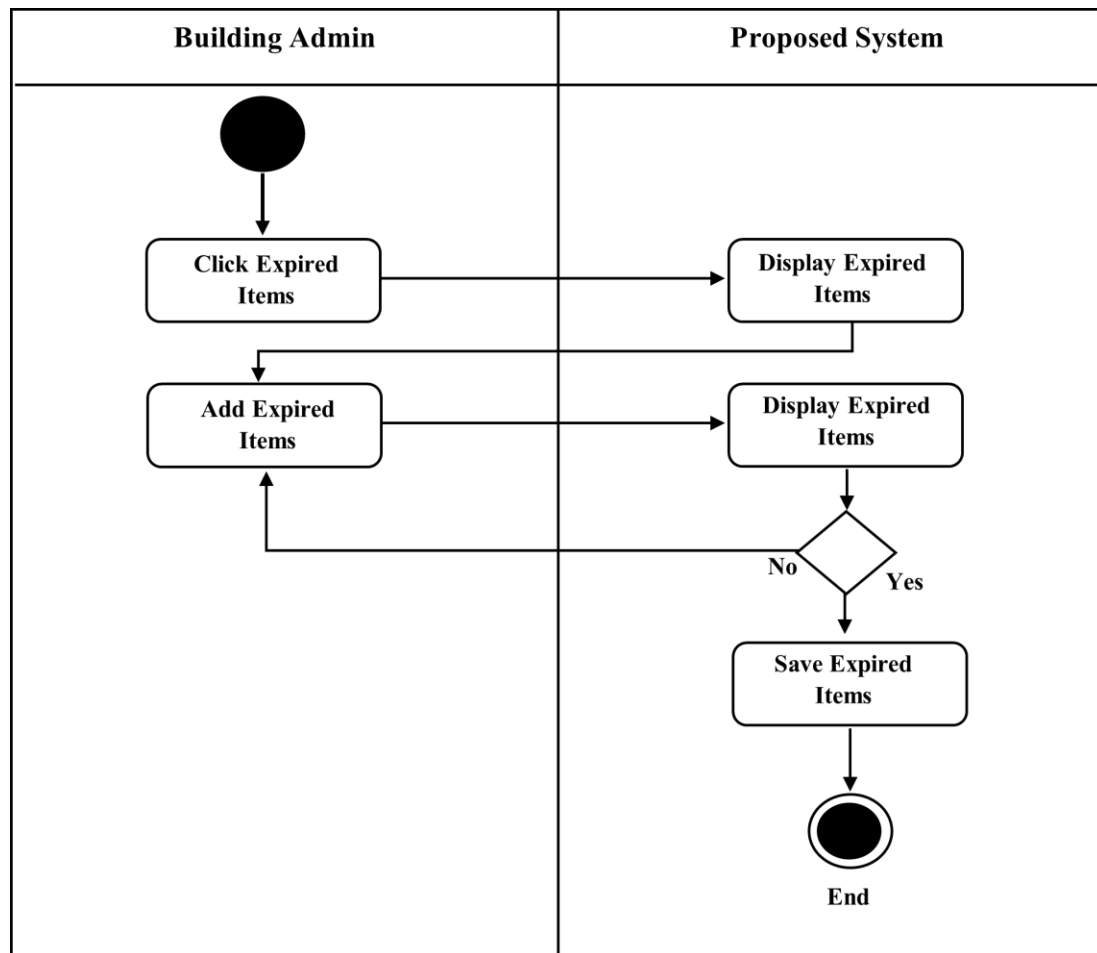


Figure 41: Activity Diagram (Expired Items)

Figure 41 Activity Diagram (Expired Items) shows the process of adding expired items to the system and how it gets stored to the systems database.

3.7 ENTITY RELATIONSHIP DIAGRAM

An entity–relationship model (ER model) is a data model for describing the data or information aspects of a business domain or its process requirements, in an abstract way that lends itself to ultimately being implemented in a database such as a relational database. The main components of ER models are entities and the relationships that can exist among them, and databases.

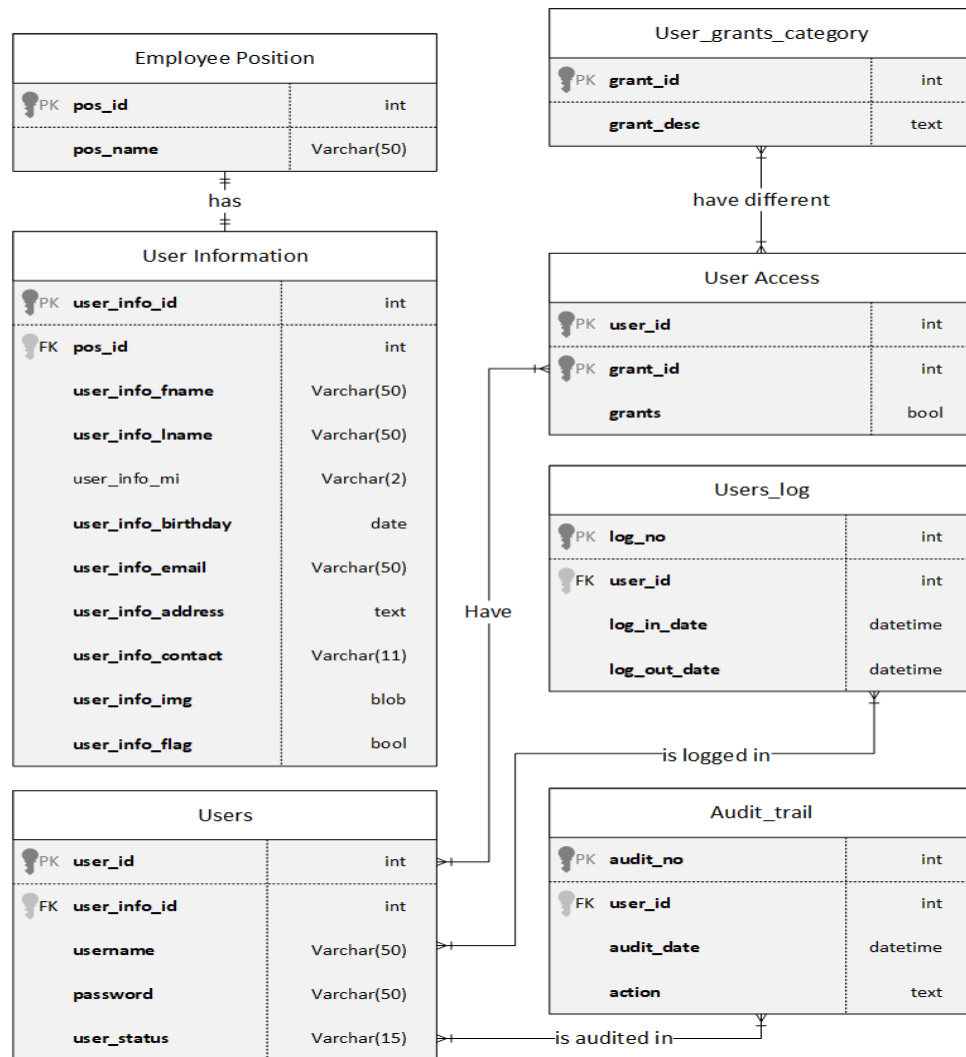


Figure 42: Entity Relationship Diagram

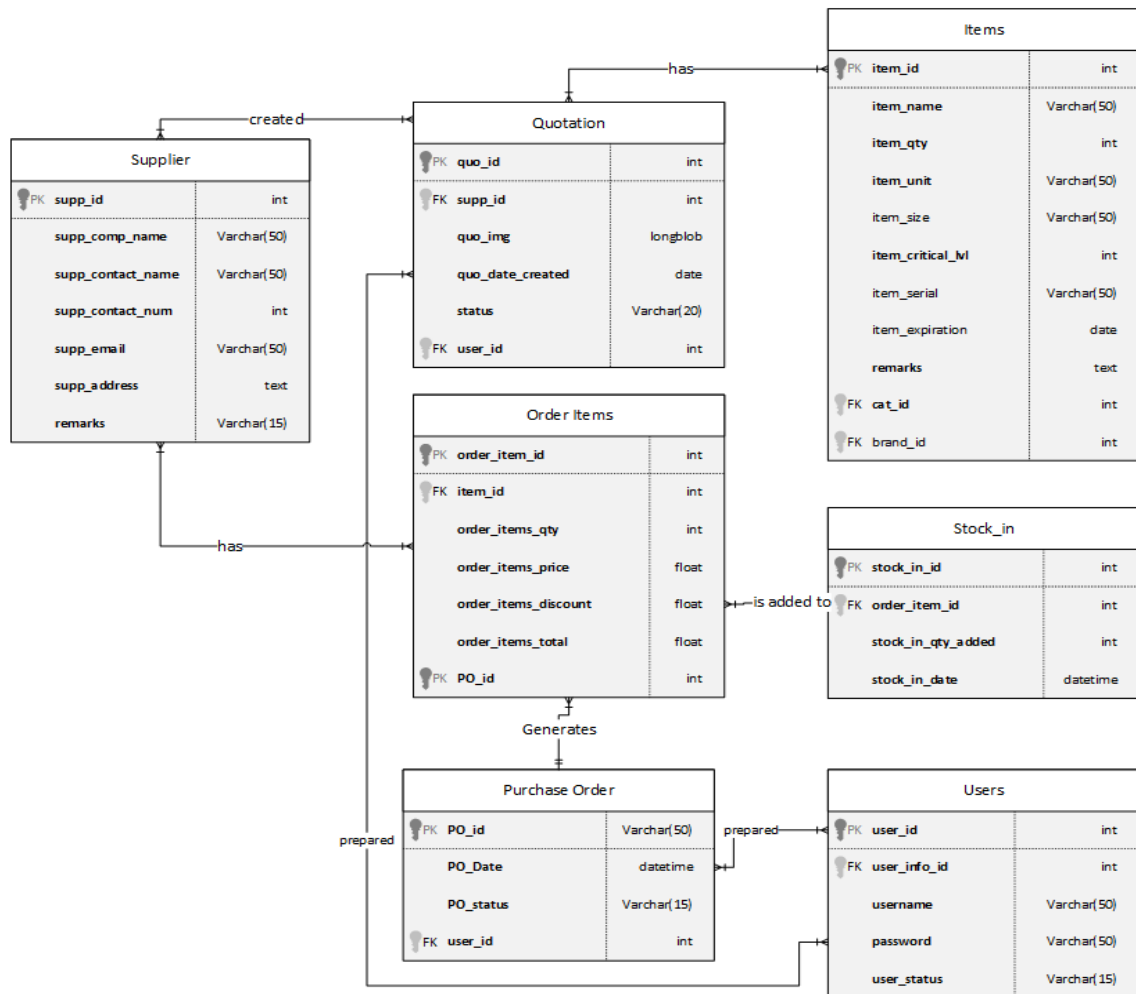


Figure 43: Entity Relationship Diagram

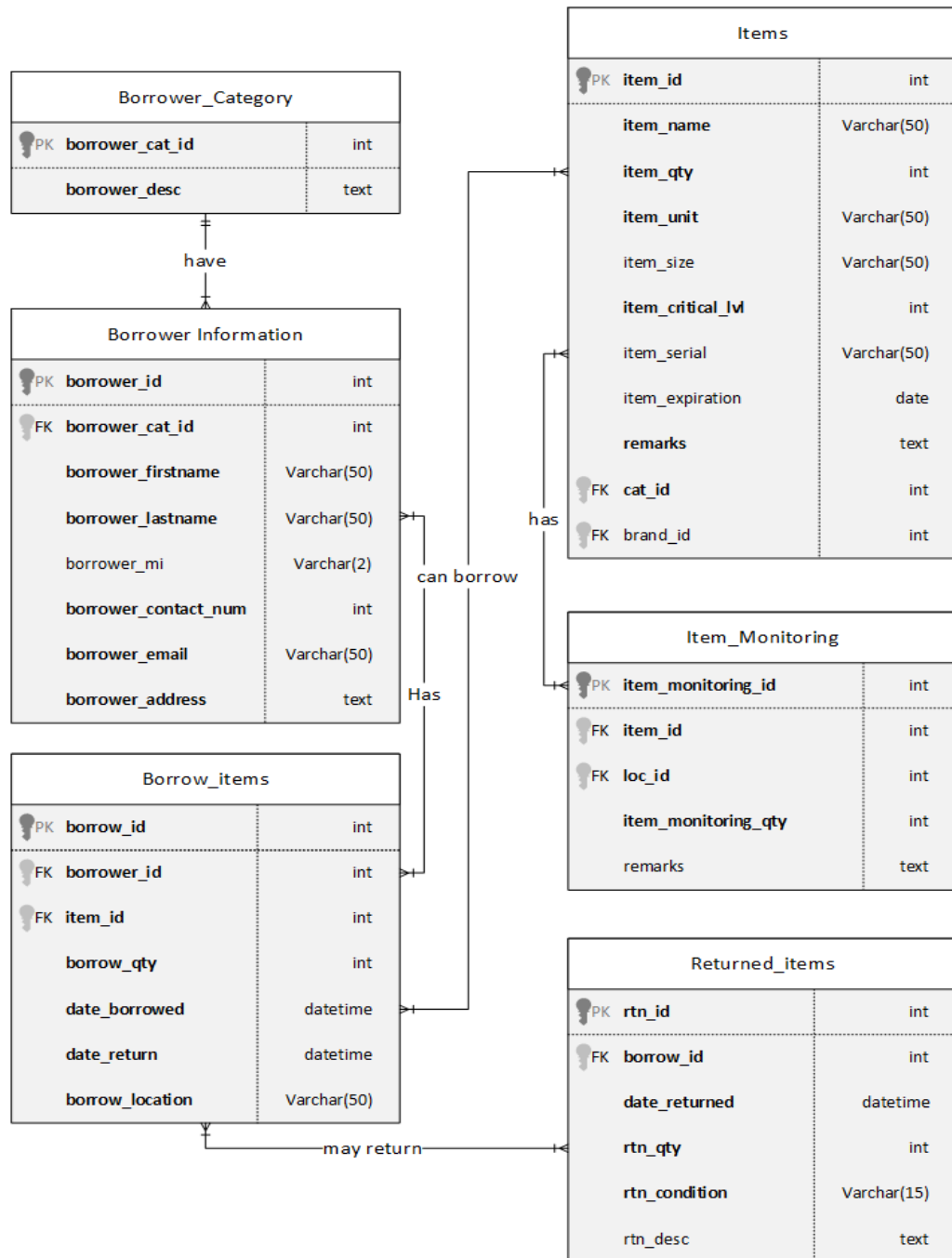


Figure 44: Entity Relationship Diagram

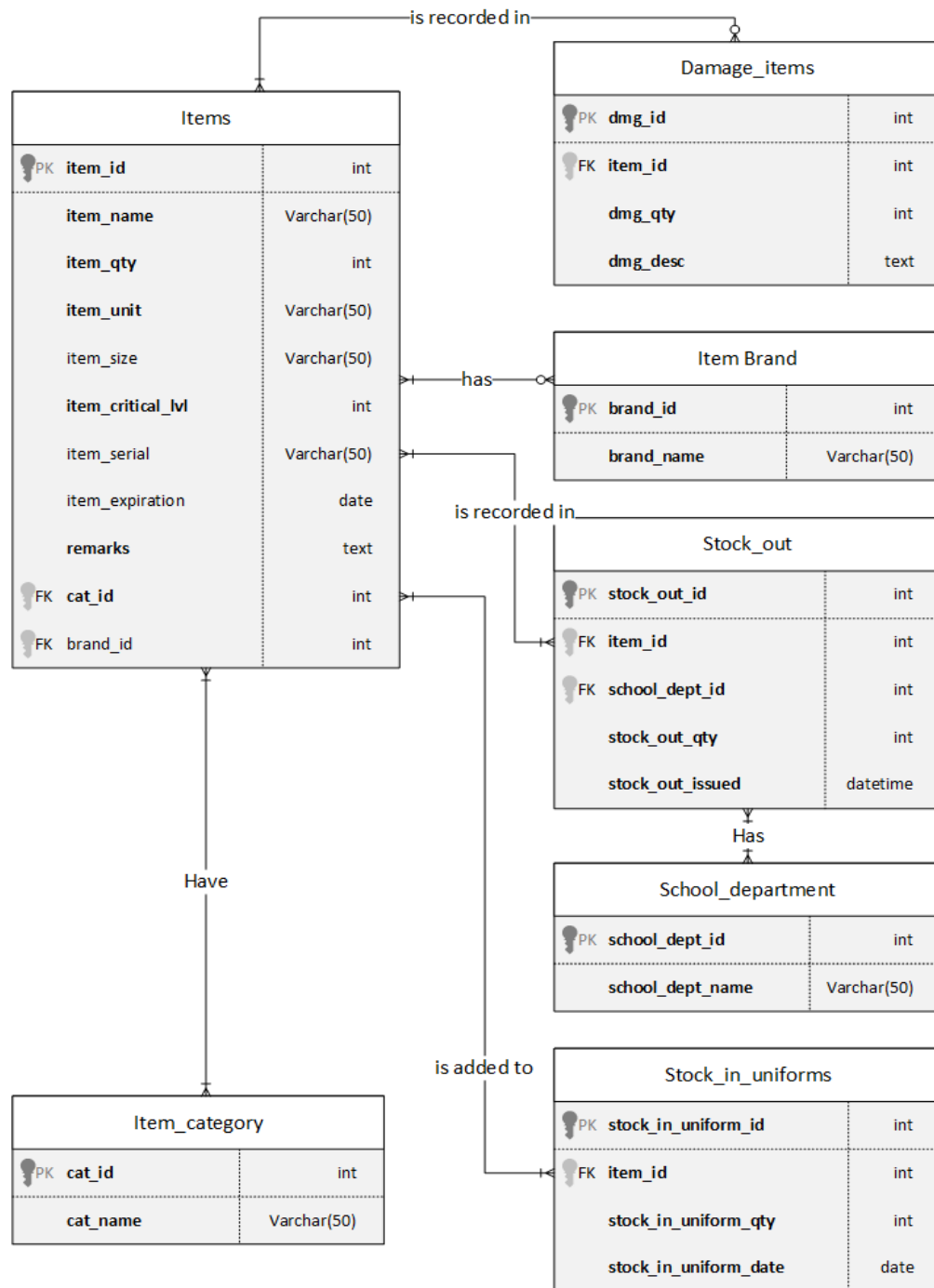


Figure 45: Entity Relationship Diagram

The entity relationship diagram shows the flow and relationships of the proposed system. It will start when the department head adds a series of information to the system, like user information, user access, item brands, location, school department, items, supplier, and borrower information. Once the department head adds any of the mentioned information to the system, it will be automatically saved to the respective tables in the database. Each entity mentioned above would typically correspond to a table in the database, and the attributes would represent the columns in those tables. Additionally, the system stores all other transactions and movements in the database audit trail and user log. These logs capture the history of user actions, system events, and any relevant information for tracking and auditing purposes. The informations related to quotations, purchase orders, and order items would be stored in their respective tables in the database. These tables would store the relevant details for each quotation, purchase order, and order item, along with any additional attributes required. After the quotation and purchase order process is done and if the items are already delivered the admins can now update and their respective inventories. The information related to inventory items, stock-in processes, and stock-out processes would be stored in their respective tables in the database. The stock-in processes would update the inventory item quantity by adding the received items to the existing stock. Conversely, the stock-out processes would reduce the inventory item quantity by deducting the items that have been taken out.

3.8 USE CASE

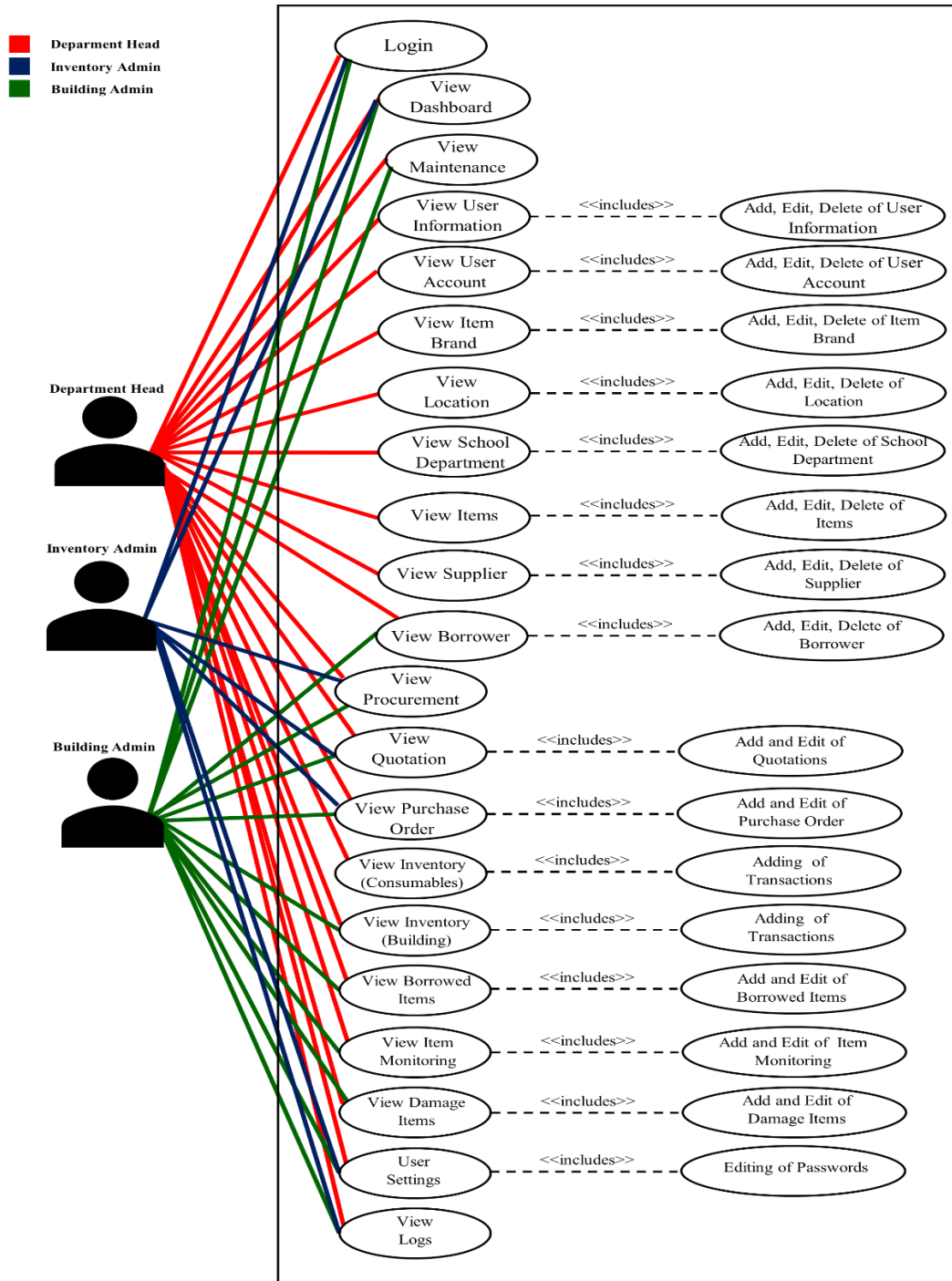


Figure 46 : Use Case of the Proposed System

Figure 46 Use Case of the Proposed System represents the access rights of the department head. All processes can be accessed by the department head. With the department head having access to all processes, it implies he can perform a wide range of actions or functionalities within the system. This include adding user information, managing user access, adding item brands, managing location and school department information, creating and managing purchase orders, managing inventory, and other relevant tasks associated with their role.

The access rights of the building admin. However, the access of building admin to other modules or functionalities may be limited but in this case, the building admin is only allowed to click on the borrower module, suggesting that he can view, add, edit, and delete information related to borrowers, but he doesn't have full control or permission to perform all actions within the maintenance module. The building admin can access the procurement module, which typically involves activities related to quotation and purchasing of inventory items. Within this module, the building admin can be able to perform tasks such as creating quotations and purchase orders. The building admin can also change his password in the user settings and view his logs.

Unlike the building admin the only module inventory admin can access are the procurement and inventory module. The inventory admin can also access the procurement module, which typically involves activities related to quotation and purchasing of inventory items. Within this module, the inventory admin can be able to perform tasks such as creating quotations and purchase orders. While in the inventory module the admin can do transactions. This indicates that the inventory admin has the authority to fulfill and process supply requests made by other users or departments. Once a supply request is approved, the inventory admin can initiate the process of releasing the requested supplies. After the transaction has been made the system will now update the stocks in the inventory. The inventory admin can also change his password in the user settings and view his logs.

3.9 DATA DICTIONARY (ANALYSIS)

These tables present the tables used in the system and defines what data should be stored in each table.

Table 3: User Information

Field Name	Description	Field Type	Field Size	Type of Key
user_info_id	id of user information	int	--	Primary Key
pos_id	position id of the user	int	--	Foreign Key
user_info_fname	first name of the user	Varchar	50	--
user_info_lname	last name of the user	Varchar	50	--
user_info_mi	middle name of the user	Varchar	2	--
user_info_birthday	birthdate of the user	date	--	--
user_info_email	user's email address	Varchar	50	--
user_info_address	user's house address	text	--	--
user_info_contact	user's contact	Varchar	11	--
user_info_img	image of the user	blob	--	--
user_info_flag	for those user who doesn't have an account.	bool	--	--

Table 3 User Information: represents the data stored in User Information table. This includes the users basic information like fullname, birthday, email address, home address, and contact information.

Table 4: Users

Field Name	Description	Field Type	Field Size	Type of Key
user_id	id of user per registration	int	--	Primary Key
user_info_id	id of user information	int	--	Foreign Key
username	username of the user	Varchar	50	--
password	password of the user	Varchar	50	--
user_status	status of the user	Varchar	15	--

Table 5 Users represents the data stored in Users table. This table includes the login credentials of the user as well as the user id and user status.

Table 5: Employee Position

Field Name	Description	Field Type	Field Size	Type of Key
pos_id	position id of the user	int	--	Primary Key
pos_name	position name of the user	Varchar	50	--

Table 4 Employee Position represents the data stored in Employee Position table. This includes the position ID and name of the user.

Table 6: User_grants_category

Field Name	Description	Field Type	Field Size	Type of Key
grant_id	id of user granting	int	--	Primary Key
grant_desc	description of user granting	text	50	--

Table 6 Users_grants_category represents the data stored in Users grants table. This table includes the grant id and description of the user.

Table 7: User_Access

Field Name	Description	Field Type	Field Size	Type of Key
user_id	id of user per registration	int	--	Primary Key
grant_id	id of user granting	int	--	Primary Key
grants	user's grant	bool	--	--

Table 7 Users_Access represents the data stored in Users Access table. This table gives the user's id, grants, and access rights to the system.

Table 8: Item_Brand

Field Name	Description	Field Type	Field Size	Type of Key
brand_id	id per item's brand	int	--	Primary Key
brand_name	brand name of the item	Varchar	50	--

Table 8 Item_Brand represents the data stored in Item Brand table. This includes the id and name of the brand.

Table 9: Items

Field Name	Description	Field Type	Field Size	Type of Key
item_id	id of the item	int	--	Primary Key
item_name	item name	Varchar	50	--
item_qty	item quantity	int	--	--
item_unit	item unit	Varchar	50	--
item_size	item size	Varchar	50	--
item_critical_lvl	items on critical levels	int	--	--
item_serial	item's serial number	Varchar	50	--
item_expiration	expiration date of items	date	--	--
remarks	item remarks	text	--	--
cat_id	category id	int	--	Foreign Key
brand_id	id per item's brand	int	--	Foreign Key

Table 9 Items represents the data stored in Items table. This includes the items id, name, quantity, unit, sizes, serial number, expiration date as well as the id of category and item brand.

Table 10: Item_location

Field Name	Description	Field Type	Field Size	Type of Key
loc_id	location id of the item	int	--	Primary Key
loc_target_qty	number of items per location	int	--	--
loc_name	location name of the item	Varchar	50	--

Table 10 Item_location represents the data stored in Item location table. This includes the location id, name, and number of items per location.

Table 11: School_department

Field Name	Description	Field Type	Field Size	Type of Key
school_dept_id	id per school department	int	--	Primary Key
school_dept_name	name of the school department	Varchar	50	--

Table 11 School_department represents the data stored in School department table. This includes the school department's id and name.

Table 12: Supplier

Field Name	Description	Field Type	Field Size	Type of Key
supp_id	supplier's id	int	--	Primary Key
supp_comp_name	company name of the supplier	Varchar	50	--
supp_contact_name	contact person of the supplier	Varchar	50	--
supp_contact_num	supplier's contact number	int	--	--
supp_email	supplier's email address	Varchar	50	--
supp_address	supplier's company address	text	--	--
remarks	availability of the supplier	Varchar	15	--

Table 12 Supplier represents the data stored in Supplier table. This includes the supplier's id, company name, company's contact person and number, suppliers email address, company address, and the availability of the supplier.

Table 13: Borrower_information

Field Name	Description	Field Type	Field Size	Type of Key
borrower_id	id of borrower per borrow	int	--	Primary Key
borrower_cat_id	description of the borrower	int	--	Foreign Key
borrower_firstname	first name of the borrower	Varchar	50	--
borrower_lastname	last name of the borrower	Varchar	50	--
borrower_mi	middle name of the borrower	Varchar	2	--
borrower_contact_num	borrower's phone number	int	-	--
borrower_email	borrower's email address	Varchar	50	--
borrower_address	borrower's home address	text	---	--

Table 13 Borrower_information represents the data stored in Borrower information table. This includes the id, category id, borrowers basic information, contact information, as well as the borrowers email address and home address.

Table 14: Borrower_Category

Field Name	Description	Field Type	Field Size	Type of Key
borrower_cat_id	borrower position number	int	--	Primary Key
borrower_desc	borrower description	text	--	--

Table 14 Borrower_Category represents the data stored in Borrower Category table. This includes the id, category id, borrowers basic information, contact information, as well as the borrowers email address and home address.

Table 15: Item_category

Field Name	Description	Field Type	Field Size	Type of Key
cat_id	category id	int	--	Primary Key
cat_name	category name	Varchar	50	--

Table 15 Item_Category represents the data stored in Item Category table. This includes the category id and name of the item.

Table 16: Quotation

Field Name	Description	Field Type	Field Size	Type of Key
quo_id	id per quotation	int	--	Primary Key
supp_id	suppliers id	int	--	Foreign Key
quo_img	image of the quotation	longblob	--	--
quo_date_created	creation date of the quotation	date	--	--
status	status of the quotation	Varchar	20	--
user_id	id of the user	--		Foreign Key

Table 16 Quotation represents the data stored in Quotation table. This includes the quotation id, suppliers id, quotation image, date created, the status of the quotation, and who made that quotation.

Table 17: Purchase Order

Field Name	Description	Field Type	Field Size	Type of Key
PO_id	purchase order id	Varchar	50	Primary Key
PO_date	purchase order date	datetime	--	--
PO_status	status of the purchase order	Varchar	15	--
user_id	user id of the personnel who make the purchase order	int	--	Foreign Key

Table 17 Purchase Order represents the data stored in Purchase Order table. This includes the purchase order id, date, the status of the purchase order, and who made that purchase order.

Table 18: Order_items

Field Name	Description	Field Type	Field Size	Type of Key
order_item_id	id of item per order	int	--	Primary Key
items_id	id of the item	int	--	Foreign Key
order_items_qty	quantity of items ordered	int	--	--
order_items_price	price of the item ordered	float	--	--
order_items_discount	discounted item	float	--	--
order_items_total	total price of the ordered items	float	--	--
PO_id	purchase order id	int	--	Primary Key

Table 18 Order_items represents the data stored in Order items table. This includes the necessary details needed like the order item id, item id, quantity of order items, price, discount, and total as well as the purchase order id.

Table 19: Stock_in

Field Name	Description	Field Type	Field Size	Type of Key
stock_in_id	id per item restocking	int	--	Primary Key
order_item_id	id of item per order	int	--	Foreign Key
stock_in_qty_added	number of items added	int	--	--
stock_in_date	date of restocking	datetime	--	--

Table 19 Stock_in represents the data stored in Stock in table. This includes the id per stock in, order item id, quantity of stocks added, and date of restocking.

Table 20: Stock_in_uniforms

Field Name	Description	Field Type	Field Size	Type of Key
stock_in_uniform_id	id of uniform per restocking	int	--	Primary Key
item_id	id of the item	int	--	Foreign Key
stock_in_uniform_qty	quantity of added uniform	int	--	--
stock_in_uniform_date	date of uniform adding	date	--	--

Table 20 Stock_in_uniforms represents the data stored in Stock in uniforms table. This includes the id of uniform per restocking, item id, quantity added, and date of restocking.

Table 21: Item_Monitoring

Field Name	Description	Field Type	Field Size	Type of Key
item_monitoring_id	id of item per monitoring	int	--	Primary Key
item_id	id of the item	int	--	Foreign Key
loc_id	location id of the item	int	--	Foreign Key
item_monitoring_qty	number of item per location	int	--	--
remarks	remarks of the monitored items	text	--	--

Table 21 Item_Monitoring represents the data stored in Item Monitoring table. This includes the monitoring id, item id, location id, number of items per location, and remarks on the monitored items.

Table 22: Damage_items

Field Name	Description	Field Type	Field Size	Type of Key
dmg_id	id of damage items	int	--	Primary Key
item_id	id of the item	int	--	Foreign Key
dmg_qty	quantity of damage items	int	--	--
dmg_desc	description of damage items	text	--	--

Table 22 Damage_items represents the data stored in Damage items table. This includes the id per damage items, item id, quantity of the damage items, and damage item description.

Table 23: Stock_out

Field Name	Description	Field Type	Field Size	Type of Key
stock_out_id	id per item withdrawal	int	--	Primary Key
item_id	id of the item	int	--	Foreign Key
school_dept_id	school department id	int	--	Foreign Key
stock_out_qty	quantity of item pers withdrawal	int	--	--
stock_out_issued	date of item withdrawal	datetime	--	--

Table 23 Stock_out represents the data stored in Stoc table. This includes the monitoring id, item id, location id, number of items per location, and remarks on the monitored items.

Table 24: Borrow_items

Field Name	Description	Field Type	Field Size	Type of Key
borrow_id	transaction id per borrower	int	--	Primary Key
borrower_id	id of borrower per borrow	int	--	Foreign Key
item_id	id of the item borrowed	int	--	Foreign Key
borrow_qty	number of borrowed	int	--	--
date_borrowed	borrowing date	datetime	--	--
date_return	return date of the borrowed item	datetime	--	--
borrow_location	location of the borrowed items	Varchar	50	--

Table 24 Borrow_items represents the data stored in Borrowed items table. This includes the borrow id, borrowers id, item id, quantity of borrowed items, date borrowed, return date of the borrowed items, and the location of the borrowed items.

Table 25: Returned_items

Field Name	Description	Field Type	Field Size	Type of Key
rtn_id	id per item return	int	--	Primary Key
borrow_id	transaction id per borrower	int	--	Foreign Key
date_returned	date of the item returned	datetime	--	--
rtn_qty	quantity of return item	int	--	--
rtn_condition	condition of return item	Varchar	15	--
rtn_desc	return item description	text	--	--

Table 25 Returned_items represents the data stored in Returned items table. This includes the return id, borrow id, returned date of the item, quantity of return id, condition of item returnm and return description.

Table 26: Audit_trail

Field Name	Description	Field Type	Field Size	Type of Key
audit_no	id of transactions on the system	int	--	Primary Key
user_id	id of user	int	--	Foreign Key
audit_date	date of transactions on the system	datetime	--	--
action	actions user made on the system	text	--	--

Table 26 Audit_trail represents the data stored in Audit trail table. This includes the audit number, user id, date of transactions, and actions made.

Table 27: Users_Log

Field Name	Description	Field Type	Field Size	Type of Key
log_no	login id of the user	int	--	Primary Key
user_id	id of user	int	--	Foreign Key
log_in_date	login date	datetime	--	--
log_out_date	logout date	datetime	--	--

Table 27 Users_Log represents the data stored in Users Log table. This includes the login id, id of the user, as well as the login and logout date of ther user.

3.10 ROLES AND RESPONSIBILITIES

MEMBERS	ROLES	RESPONSIBILITIES
 Ronald M. Bantayan	Project Manager Quality Assurance	- Leads the group - Consolidate all the reports - One of the system tester (QA)
 Armando A. Calma Jr.	Senior Programmer Database Administrator	- In charge of writing codes and developing the system - Designed the database - Responsible for sending regular updates to the Project Manager.

 Mark Joseph R. Accad	Documentation Quality Assurance	- One of the system tester (QA) - Leads the documentation - In charge of making all the presentations needed.
 Kenneth E. Maquiñana	Graphic Designer	- In charge of graphic designing - Logo designing - System Interface Planning.

Table 28: Roles and Responsibilities

Table 28 Roles and Responsibilities represents the proponents role and responsibilities on this project.